TYRX 可吸收抗菌包膜

The TYRX Absorbable Antibacterial Envelope

TY: Tyrosine 酪胺酸聚合物

Rx: Minocycline + Rifampin 兩種抗生素

謝慕揚 MD, PhD, FESC

譯自 Medtronic「TYRX 2026 Overview Deck」| 2026-10-17

大綱

1. 產品原理：裝置穩定與囊袋癒合 (pocket healing)
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4. 禁忌症與注意事項
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> 本講義為教學翻譯整理；原始投影片屬 Medtronic 行銷素材，數據請以原始文獻為準。每頁註解與標題中的試驗名稱都是可點選的全文連結。

一、產品原理

裝置穩定 (stabilization) 與囊袋癒合 (pocket healing)

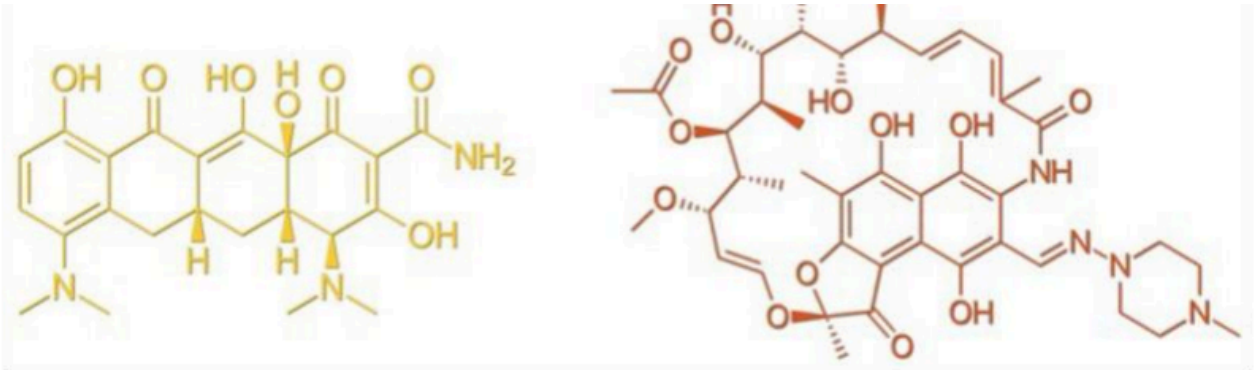
TYRX 包膜：三個核心設計

- **聚合物控制釋放**：抗生素由 tyrosine 聚合物控制溶出，至少維持 7 天
- **局部給藥**：minocycline 與 rifampin 直接釋放到囊袋組織
- **可吸收網布**：多股編織網布，約 9 週吸收

右圖：植入後、4 週、約 9 週的包膜外觀變化。

來源：Medtronic 臨床前研究 (Huntingdon Life Sciences TR-2013-001；Sinclair Labs D13599)，為 data on file，無公開連結。

兩種抗生素的化學結構



Minocycline　　Rifampin

- **Minocycline**：tetracycline 類，抑菌 (bacteriostatic)，抑制蛋白質合成
- **Rifampin**：殺菌 (bactericidal)，抑制 DNA-dependent RNA polymerase
- 兩者合用對 coagulase (-) staphylococci 與 S. aureus 有互補作用

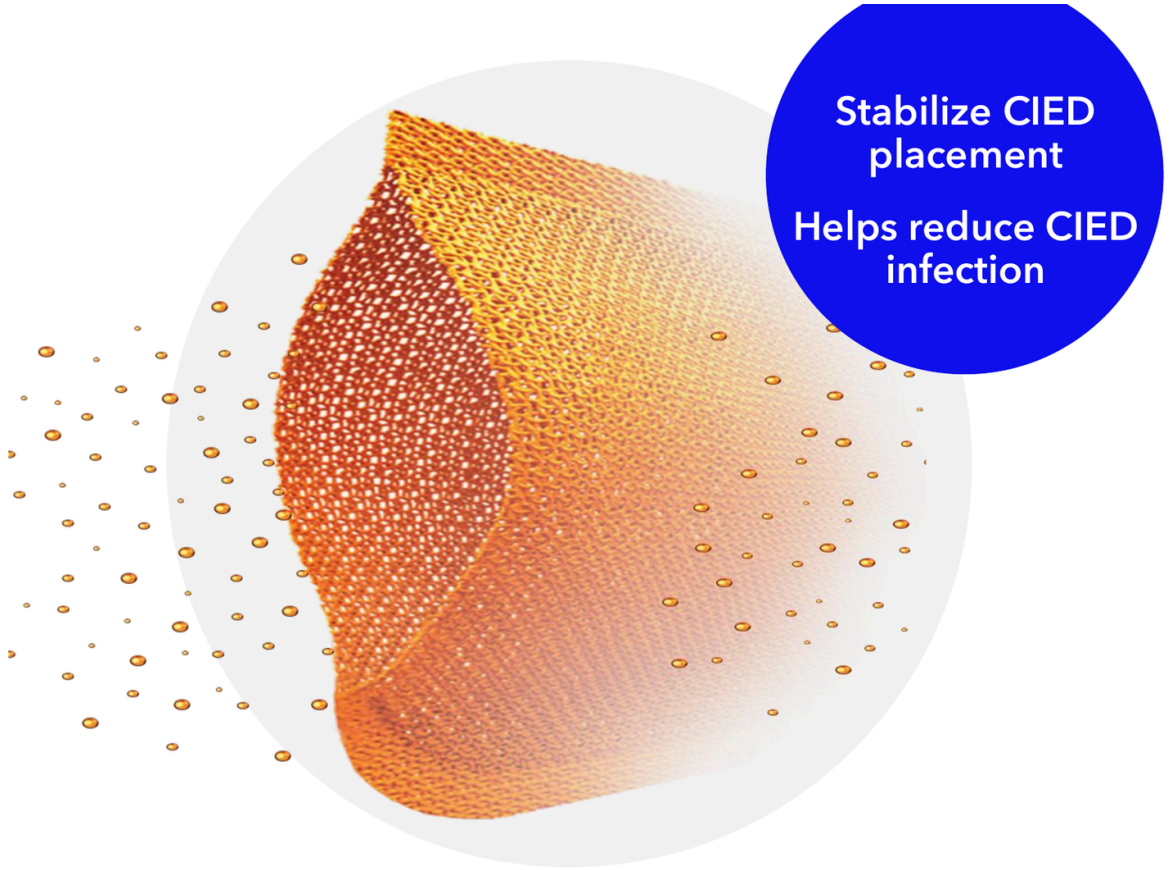
體外協同作用：Zinner SH, et al. J Infect Dis 1981;144:365-371；Segreti J, et al. Diagn Microbiol Infect Dis 1989;12:253-255。

四個理由：T-I-M-E

- **T**widdler syndrome：減少裝置被纏繞
- **I**nfection：降低 CIED 感染
- **M**igration：減少裝置移位
- **E**rosion：減少裝置磨穿皮膚

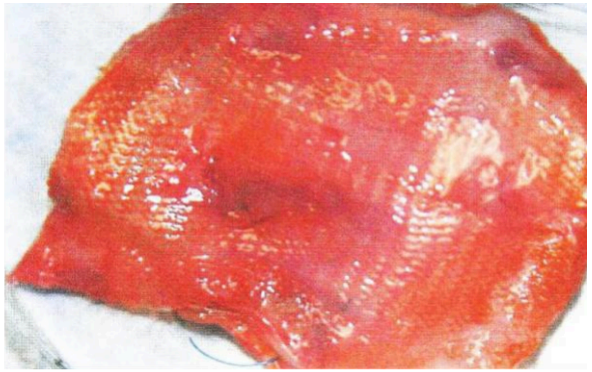
總結一句：先固定裝置，再壓低感染風險。

Twiddler 病例系列：Osoro M, et al. Pacing Clin Electrophysiol 2018;41:136-142。感染：見第五章 WRAP-IT。



;26:1111-1116.
:26:783-789.
esults: Use of Antibacterial

裝置穩定：動物實驗的組織反應



植入後 1 天	植入後 7 天	植入後 5 週
纖維結締組織反應開始	組織立即長入網布，固定裝置	網布已吸收，囊袋成形

左、中：Huntingdon Life Sciences TR-2011-054 (Medtronic data on file)。右：Francois Philippon, MD (Laval University) 提供。同類已發表的動物資料見 Virmani R, et al. Heart Rhythm 2024。

裝置穩定：為什麼網布有用

- 植入物（CIED、神經刺激器）會引發宿主反應，約 30 天形成纖維囊 (fibrous capsule)
- 活體模型顯示，TYRX 多孔網布讓緻密纖維結締組織**立即長入**，7 到 14 天內固定裝置
- 網布存在約 9 週 (63 天)，遠長於組織長入所需的時間

> 說穿了，網布只是暫時的鷹架。組織長好了，鷹架就自行消失。

囊袋組織學的人體資料：Massaro G, et al. Heart Rhythm 2023;20:198-206。

TYRX Pocket Health Study：Philippon, JCE 2025

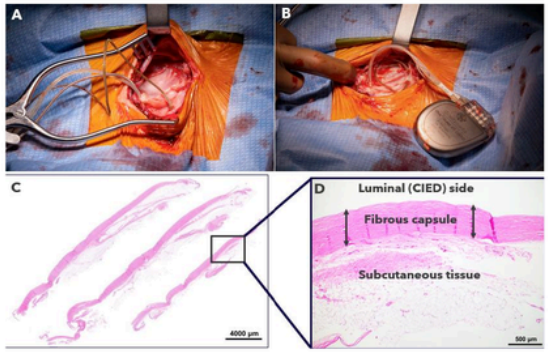
可吸收抗菌包膜促進健康的 CIED 囊袋形成

- **目的**：評估曾用 TYRX 的病人，回來做二次手術時囊袋的組織學與形態特徵
- **設計**：前瞻性觀察研究 (NCT05356546)，2022-06 至 2023-07，9 個中心
- 27 個囊袋，兩次手術平均相隔 6.8 年；與歷史對照比較

> 結果：沾黏顯著較少 ($p < 0.05$)；囊膜顯著較薄 (**0.33 ± 0.16 mm** vs 對照 0.8 ± 0.3 mm，$p < 0.0001$)，以 type I 膠原為主；96% 的囊袋發炎為無到輕度；無臨床上顯著的鈣化。

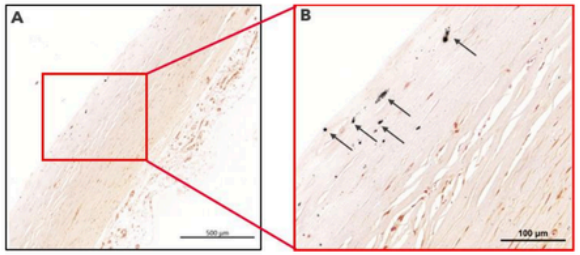
Philippon F, Ladich E, Virmani R, et al. J Cardiovasc Electrophysiol 2025;36(8)。初步結果曾於 HRS 2024 發表 (PO-04-162)。

Pocket Health Study：組織學所見

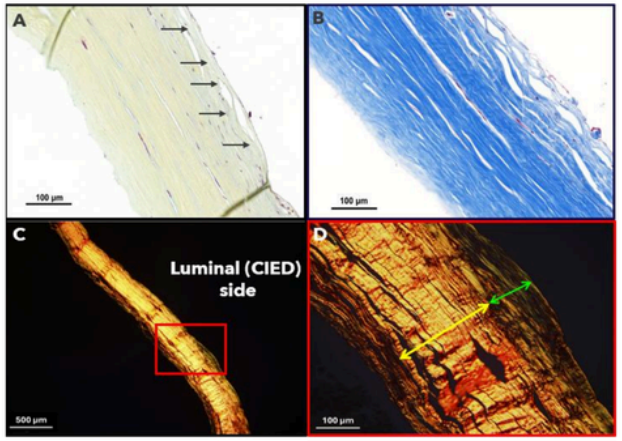


A-B) Operative photo of TYRX changeout procedure with no detectable adhesion C) Whole H&E-stained sections of the TYRX CIED pocket showing a thin band of mature collagen (double arrows; D).

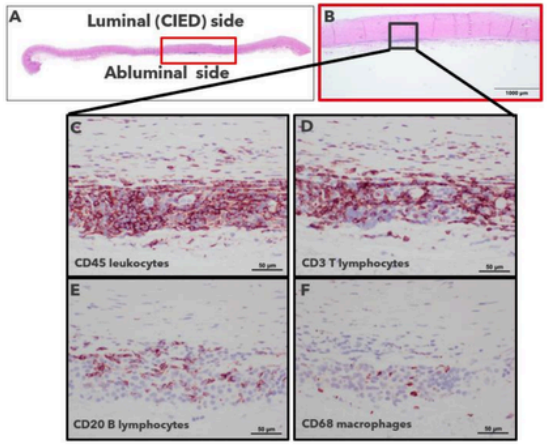
No clinically significant calcification



A) Von Kossa stain demonstrating rare occurrence of microcalcifications (B; arrows) in TYRX pocket tissue samples.



A) Capsule contains mature collage with proteoglycan deposition (arrows). B) Thin capsule with bright blue staining indicative of mature collagen. C) Compact strong pocket with type I mature collage (D; yellow arrows) and low levels of Type III collagen (D).



Whole section staining (A) if small focal inflammation (B) with a focus on chronic immune cells. Overall demonstrates low focal distribution of leukocytes (C) which are predominantly CD3-positive T lymphocytes (D) with a rare, diffuse presence of CD20-positive B lymphocytes (E) and CD68-positive macrophages (F).

* Based on study with device changeout ~7 years after previous implant

1. Philippon F, et al. Absorbable antibacterial envelope promotes development of a healthy CIED pocket. 2024 HRS conference poster presentation.

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Medtronic

左上：手術照片無沾黏，H&E 見薄層成熟膠原。中：成熟 type I 膠原。右：發炎細胞少，以 CD3 T 細胞為主。左下：Von Kossa 僅見罕見微鈣化。圖取自原簡報；全文見 Philippon F, et al. JCE 2025。

包膜對囊袋癒合的影響：Virmani, Heart Rhythm 2024

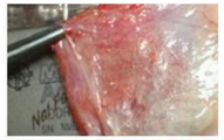
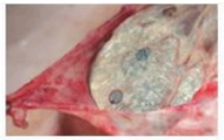
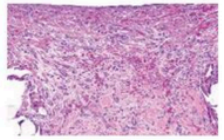
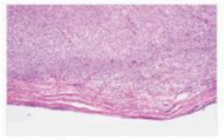
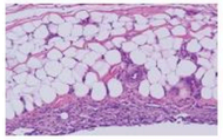
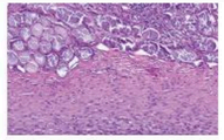
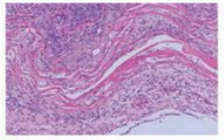
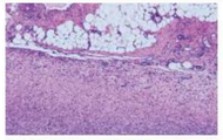
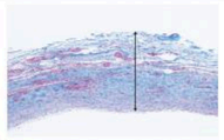
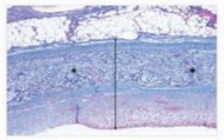
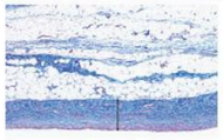
頭對頭的臨床前比較 (TYRX vs CanGaroo ECM vs 無包膜)

- **設計**：羊隻模型 (n=21)，共 110 個 CIED 囊袋
- 時間序列組織病理與形態分析，獨立盲性判讀，採 ISO 10993-6:2016 評分
- **結果**（與 CanGaroo 相比，TYRX 上市版）：
  - 組織整合更快，早期即穩定
  - 發炎更低、囊膜更薄
  - 可吸收進入自體組織

> 結論：TYRX 囊袋的發炎程度與對照組相當，provisional matrix 沉積與組織貼合更快，形成癒合良好的囊袋。

Virmani R, Philippon F, Mittal S, et al. Heart Rhythm 2024;21(8)。臨床前資料不代表臨床表現。

12 週時間序列比較

Day 3 Faster tissue integration for stability	 Accelerated provisional matrix formation	 Provisional matrix not detected	 Focal provisional matrix formation
Day 7 Lower acute inflammation marker ($p < 0.01$)‡	 Low neutrophil counts	 Elevated neutrophil counts	 Comparable to TYRX
Week 4 Lower chronic inflammation marker ($p < 0.05$)‡	 Low lymphocyte counts	 Elevated lymphocyte counts	 Comparable to TYRX
Week 12 Thinner capsule and absorption into native tissue	 0.74 mm capsule thickness (envelope absorbed into native tissue)	 1.18 mm capsule thickness (*intact ECM in fibrotic pocket wall)	 0.65 mm capsule thickness (comparable to TYRX)

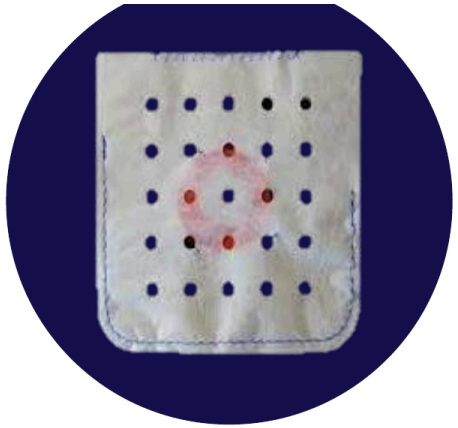
ıical data may not be indicative of clinical performance.

第 3 天：TYRX 加速 provisional matrix 形成。第 7 天與第 4 週：急性與慢性發炎指標較 ECM 低。第 12 週囊膜厚度：TYRX 0.74 mm、ECM 1.18 mm、無包膜 0.65 mm。Virmani R, et al. Heart Rhythm 2024。

二、問題的範圍

CIED 感染：發生率、後果與現行預防方式

CIED 包膜市場競爭者漸增



TYRX (Medtronic)	EluPro (Boston Scientific)	Vesta (Hylomorph)
臨床證據最完整	強調易用性與「生物性益處」	強調易用性
獲學會共識背書	搭配 BSX 合約綑綁	可吸收網布、製造成本低

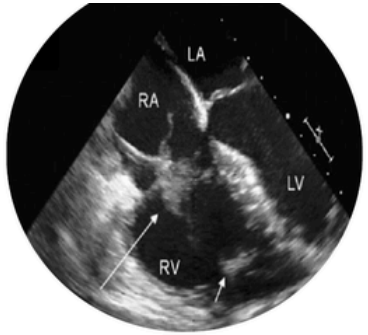
另有多種低成本非包膜選項：Tauropace、Irrisept、Collatamp、Surgicel。本頁為原簡報的市場觀點，非文獻資料。

問題的範圍

CIED 感染發生率 1 到 4%，高風險病人可達 3 到 7%

Stitch abscess

Pocket infection



Risk of

Significant morbidity

由左至右：縫線膿瘍 (stitch abscess)、囊袋感染 (pocket infection)、心內膜炎風險、顯著的併發症與死亡。

Sohail MR, et al. Pacing Clin Electrophysiol 2015;38:231-239。

挑戰與衝擊：CIED 感染

全球每年約 150 萬人接受 CIED 植入。感染是嚴重的手術相關併發症。

1 到 4%

CIED 病人發生感染

7 到 21 天

平均住院天數 (美國資料)

10 億美元

全球 CIED 感染的估計花費

22 到 56%

病人屬感染風險增加族群

> 3 倍

一年死亡風險

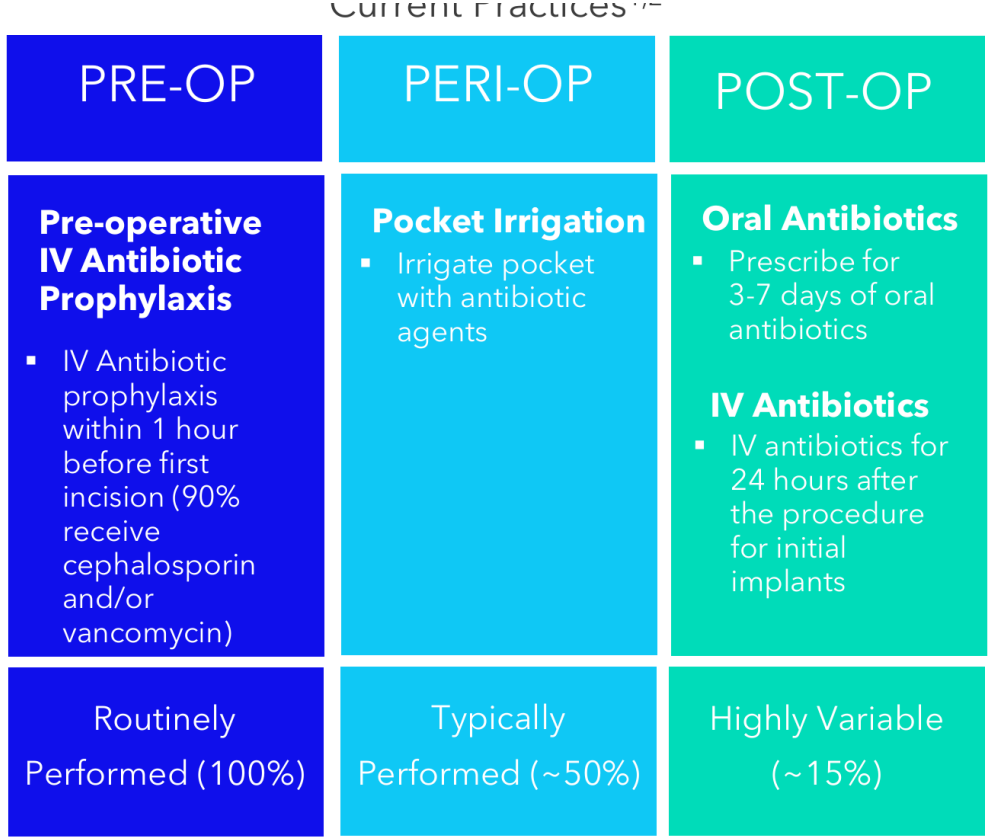
48 到 83K

美元，每例感染的平均住院費

Mond HG 2011；Tarakji KG 2016；Tarakji KG 2019；Mittal S 2014；Sohail MR 2011；Wilkoff BL 2020；Shariff N 2015。

現行 CIED 感染預防方式

- 預防方式因醫師而異，且不分風險高低，多數病人用同一套
- **術前**：切皮前 1 小時內靜脈抗生素，幾乎 100% 執行，90% 用 cephalosporin 或 vancomycin
- **術中**：囊袋抗生素沖洗，約 50% 執行
- **術後**：口服 3 到 7 天或靜脈 24 小時，差異極大，約 15%
- Cefazolin 與 vancomycin 都有缺口：**七成感染由 coagulase (-) staphylococci 與 S. aureus 造成**



執行率來自 TYRX 委託調查 (MEDACorp 2007；Easton 2008)，無公開連結。術前抗生素的 RCT 證據：de Oliveira JC, et al. Circ Arrhythm Electrophysiol 2009;2:29-34。

PADIT 試驗：加強抗生素有沒有用

Prevention of Arrhythmia Device Infection Trial

- **設計**：叢集隨機交叉試驗，4 個隨機分派的 6 個月期間
- **傳統組**：術前單劑靜脈 cefazolin
- **加強組**：術前 cefazolin 加 vancomycin、術中 bacitracin 囊袋沖洗、術後 2 天 cephalosporin
- **規模**：19,559 位病人，28 個中心，2 個國家

> 結果：傳統組感染率 1.23%，加強組 1.01%。差異不顯著。也就是，單靠全身性抗生素加碼，壓不下感染率。

Krahn AD, Longtin Y, Philippon F, et al. J Am Coll Cardiol 2018;72:3098-3109。

PADIT 感染風險分數

項目	分數
年齡 < 60 歲 / 60 到 69 歲	2 / 1
ICD / CRT / 更換或升級	2 / 4 / 4
腎功能不全 / 免疫低下	1 / 3
先前手術 1 次 / ≥ 2 次	1 / 3

- 分數 < 4：感染率 < 1%
- 分數 4 到 6：< 1.9%
- 分數 ≥ 7：< 3.5%

Birnie DH, Wang J, Alings M, et al. J Am Coll Cardiol 2019;74:2845-2854。

	OR (95% IC)	βcoefficient	P value	PADIT risk score points
Age (year) (ref. ≥70)				
<60	1.63 (1.10 – 2.41)	0.4872	0.015	2
60-69	1.43 (0.99 – 2.05)	0.3552	0.054	1
Procedure type (reference: pacemaker)				
ICD	1.83 (1.14 – 2.93)	0.6016	0.013	2
CRT	2.87 (1.83 – 4.51)	1.0547	<0.001	4
Revision/Upgrade	4.16 (2.47 – 6.32)	1.4254	<0.001	4
Renal insufficiency	1.48 (1.02 – 2.13)	0.3890	0.037	1
Immuno-compromised	2.24 (1.03 – 4.86)	0.8051	0.042	3
Number of previous procedure (ref: none)				
1	1.51 (0.98 – 2.31)	0.4114	0.059	1
≥2	3.37 (2.11-5.39)	1.2161	<0.001	3

「不是醫護人員傳給我的，是我自己的 Staph」

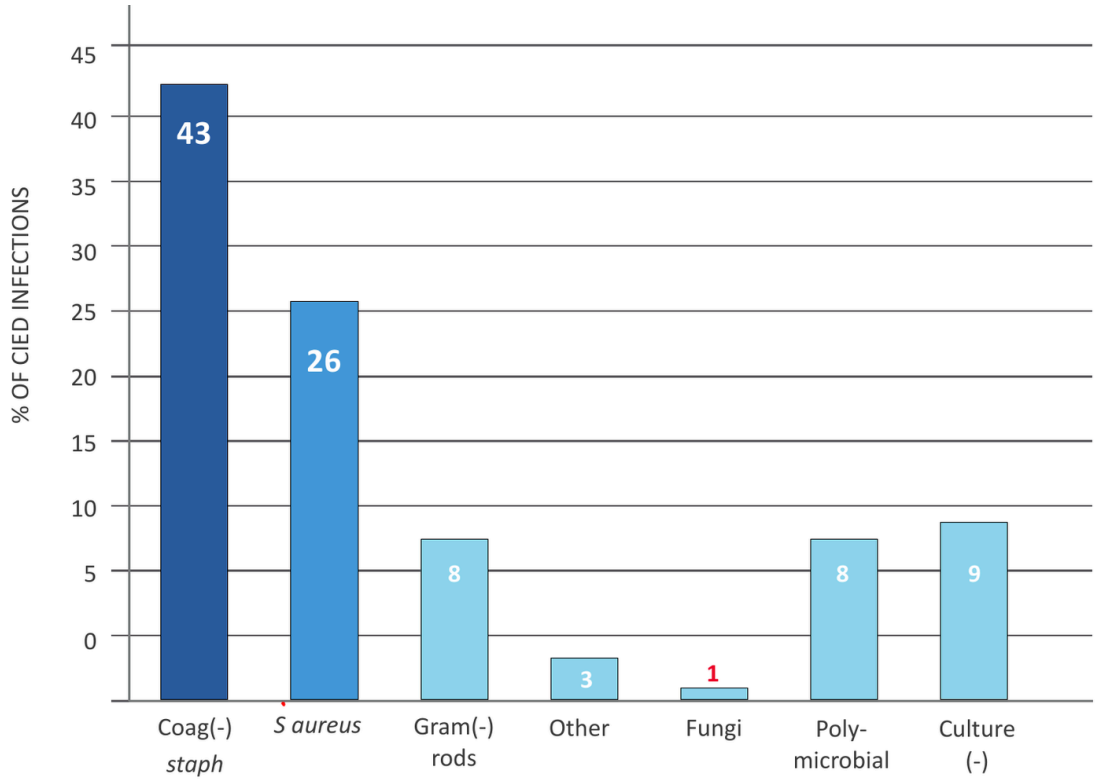
漫畫原意：病人自身的皮膚菌叢，才是 CIED 感染最主要的來源。前瞻性證據：Da Costa A, et al. Circulation 1998;97:1791-1795。

兩種細菌占約七成 CIED 感染

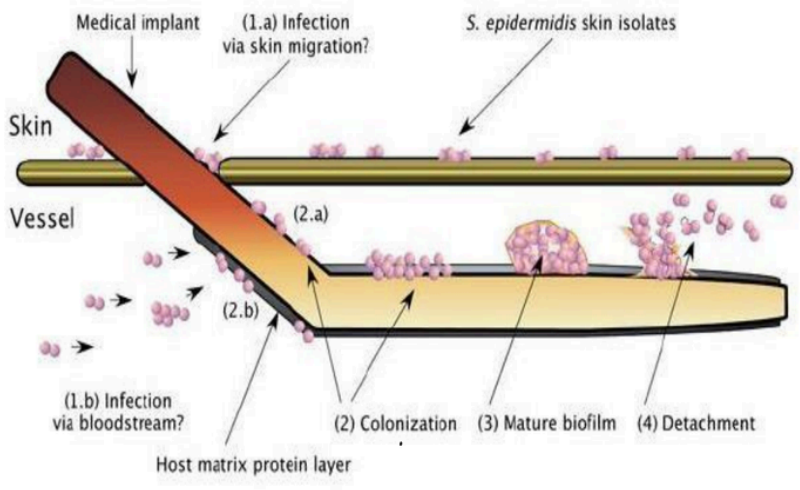
- Coagulase (-) staphylococci：**43%**
- S. aureus：**26%**
- Gram (-) rods：8%、多重菌 8%
- 培養陰性 9%、其他 3%、黴菌 1%

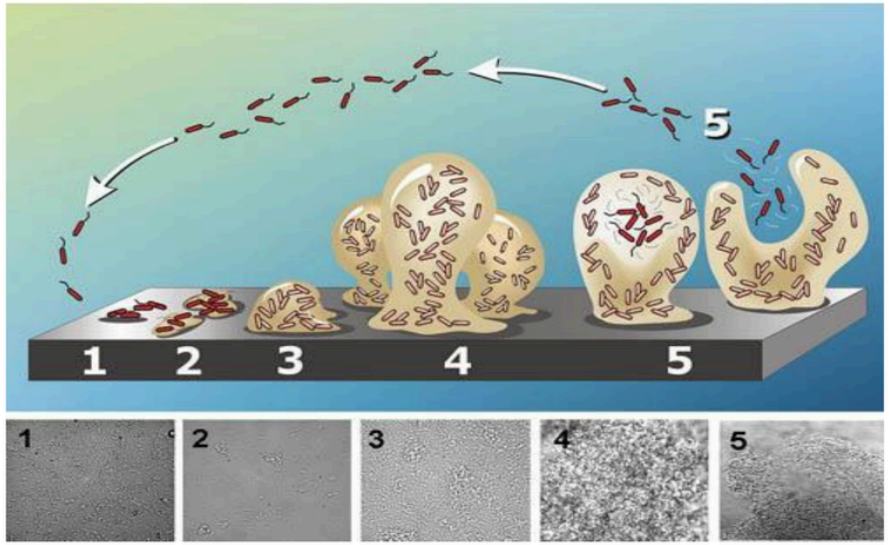
Minocycline 加 rifampin 的組合，正是針對這兩種主要致病菌。

七篇文獻彙整：Wisplinghoff 2004；Klug 2007；Da Costa 1998；de Oliveira 2009；NNIS 2004；Lekkerkerker 2009；Chua 2000。



為什麼是葡萄球菌 (staphylococci)

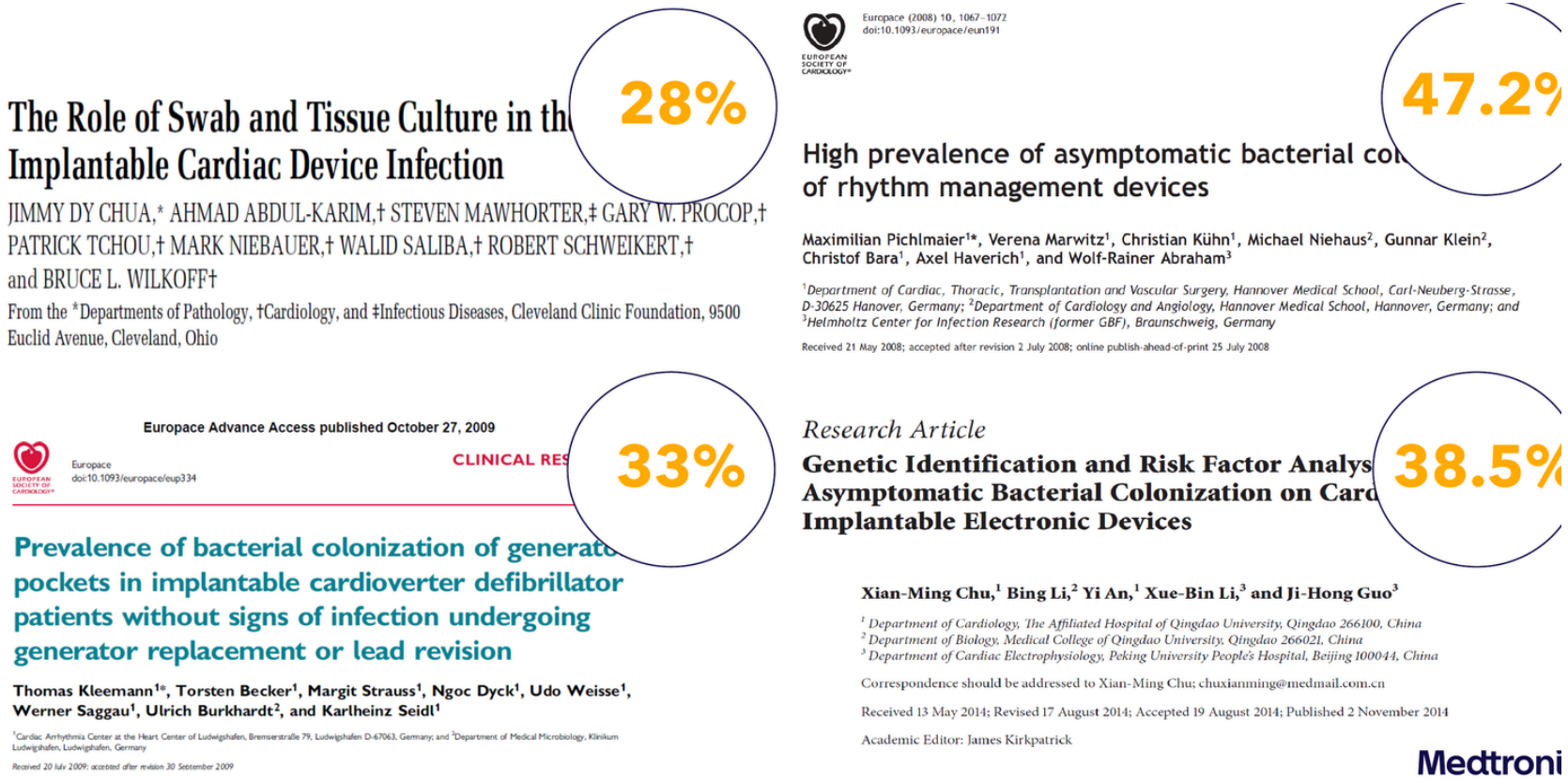




- 無所不在的**皮膚共生菌**；表現多種**黏附因子** (adherence factors)
- 會形成**生物膜** (biofilm)
- 因此是裝置相關血流感染的最主要原因

Sohail MR, et al. J Am Coll Cardiol 2007;49:1851-1859。

囊袋細菌定殖 (colonization) 是常態



無感染徵象的病人在更換電池或導線時，囊袋培養或分子檢測陽性率：28% (Chua 2005)、33% (Kleemann 2010)、38.5% (Chu 2014)、47.2% (Pichlmaier 2008)。

打破生物膜：蟻窩的比喻

- 生物膜就像蟻窩，表面看不出來，裡面卻密密麻麻
- 全身性抗生素只能在蟻窩外灑藥
- 局部高濃度、持續釋放的抗生素，才能在細菌剛附著、還沒築窩時處理掉

> 為什麼 TYRX 重要？因為它在頭 7 天把藥直接送到細菌築窩處。

動物模型中 TYRX 消除生物膜：Hansen LK, et al. Pacing Clin Electrophysiol 2009;32:898-907。

三、藥理

為什麼是 minocycline 加 rifampin

輔助性抗生素預防：各藥物對 CIED 致病菌的涵蓋

Pathogens responsible for CIED infections	Single-agent therapy				TYRX
	Cefazolin[1]	Vancomycin[1]	Gentamicin[1]	Topical ionic silver[2-5]	Minocycline & rifampicin[1,6-8]
Coagulase (-) Staphylococcus (e.g., S epidermidis)			variable activity		
Methicillin-sensitive S aureus (MSSA)			variable activity		
Methicillin-resistant S aureus (MRSA)			variable activity		
E coli					
H influenzae					
M catarrhalis				no data	
Corynebacterium jeikeium				no data	

深色格代表有活性；淺色格代表活性不一；空白代表無活性或無資料。TYRX 的 minocycline 加 rifampin 對表列所有菌種皆有涵蓋。

Sanford Guide；Percival SL 2011；Zinner SH 1981；Darouiche RO 1995；Segreti J 1989。

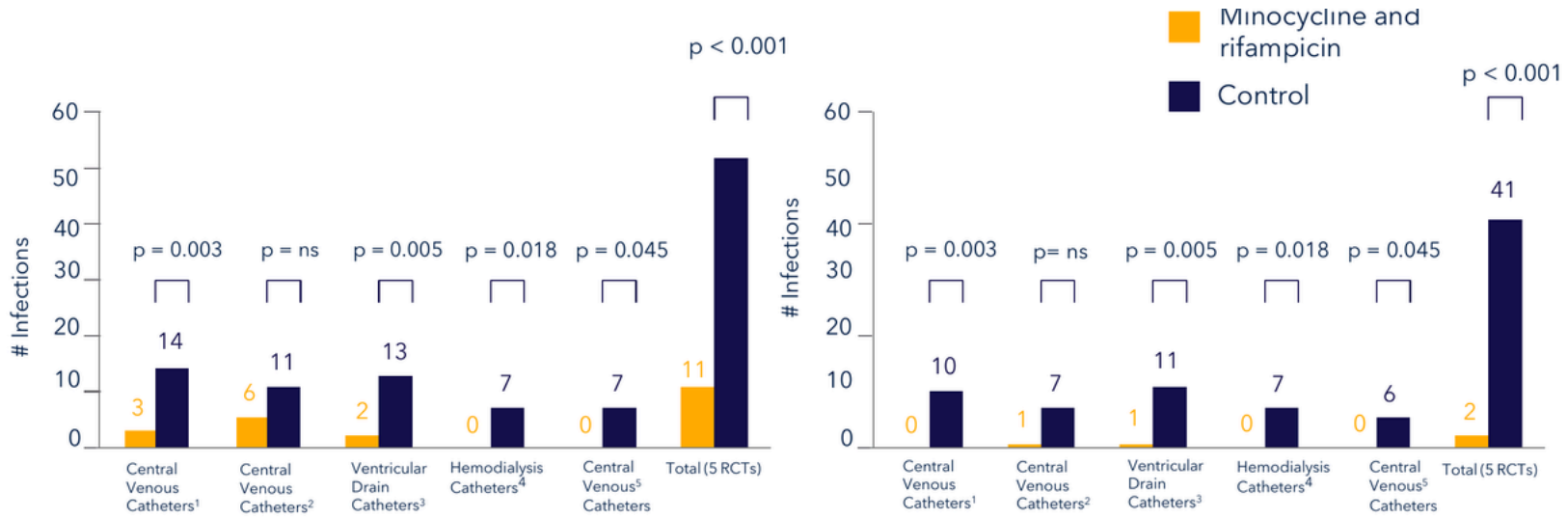
單一藥物的缺口

- **Cefazolin 與 vancomycin** 很少合併使用，單獨用來預防 CIED 感染有明顯不足
  - 兩者涵蓋範圍大量重疊，都針對 gram (+)
  - 對 gram (-) 都不強
- **Gentamicin** 對 coagulase (-) staphylococci、MSSA、MRSA 活性不一，不可靠
- **外用離子銀** 對 coagulase (-) staphylococci 沒有活性，對 M. catarrhalis 與 Corynebacterium jeikeium 沒有資料

> 簡單來說，現行預防用藥沒有一種同時穩定涵蓋最常見的兩種致病菌。

抗菌譜依據 Sanford Guide；離子銀資料 Percival SL, et al. Wound Repair Regen 2011;19:767-774。

Minocycline 加 rifampin 顯著降低感染：5 篇 RCT



- 左：整體感染，合併 5 篇 RCT 後**風險降為 1/4.8** (relative risk 0.2)
- 右：coagulase (-) staphylococci 與 S. aureus 感染，**風險降為 1/20.6** (relative risk 0.05)

導管相關感染試驗：Hanna 2004；León 2004；Zabramski 2003；Chatzinikolaou 2003；Raad 1997。

局部給藥：用 10% 的劑量達到殺菌濃度

Minocycline activity against CIED infection pathogens[1]	
Gram (+) bacteria	Gram (-) Bacteria
S aureus S pneumoniae	E coli M catarrhalis
Mechanism of action Bacteriostatic; inhibits protein synthesis	
Metabolism/excretion Liver/Renal	

Rifampicin activity against CIED infection pathogens[1]	
Gram (+) bacteria	Gram (-) Bacteria
S aureus (including MRSA) S epidermidis Cjeikeium S pneumoniae	H influenzae M catarrhalis
Mechanism of action Bacteriocidal; inhibits DNA-dependent RNA polymerase activity	
Metabolism/excretion Liver/Hepatic and Renal	

Antibiotic	Systemic (oral or IV) usual or recommended dosage	Locally delivered TYRX Absorbable Antibacterial Envelope dosage
Minocycline	200 mg initially, followed by 100 mg every 12 hours	5.1 mg (Medium) and 7.6 mg (Large)
Rifampicin	Up to 600 mg/day; 10 mg/kg, in a single daily administration	8.0 mg (Medium) and 11.9 mg (Large)

Pre-clinical *in vivo* elution studies demonstrate that the TYRX Absorbable Antibacterial Envelope generates pocket tissue minocycline

局部釋放讓囊袋組織在植入 2 小時內達到主要致病菌的 MIC，並維持至少 7 天。抗菌譜：Sanford Guide；溶出資料：Huntingdon TR-2013-001 (data on file)。

劑量比較：全身給藥 vs TYRX 包膜

抗生素	全身性 (口服或靜脈) 常用劑量	TYRX 包膜含量
Minocycline	首劑 200 mg，之後每 12 小時 100 mg	5.1 mg (Medium)、7.6 mg (Large)
Rifampin	每日最多 600 mg (10 mg/kg)，單次給予	8.0 mg (Medium)、11.9 mg (Large)

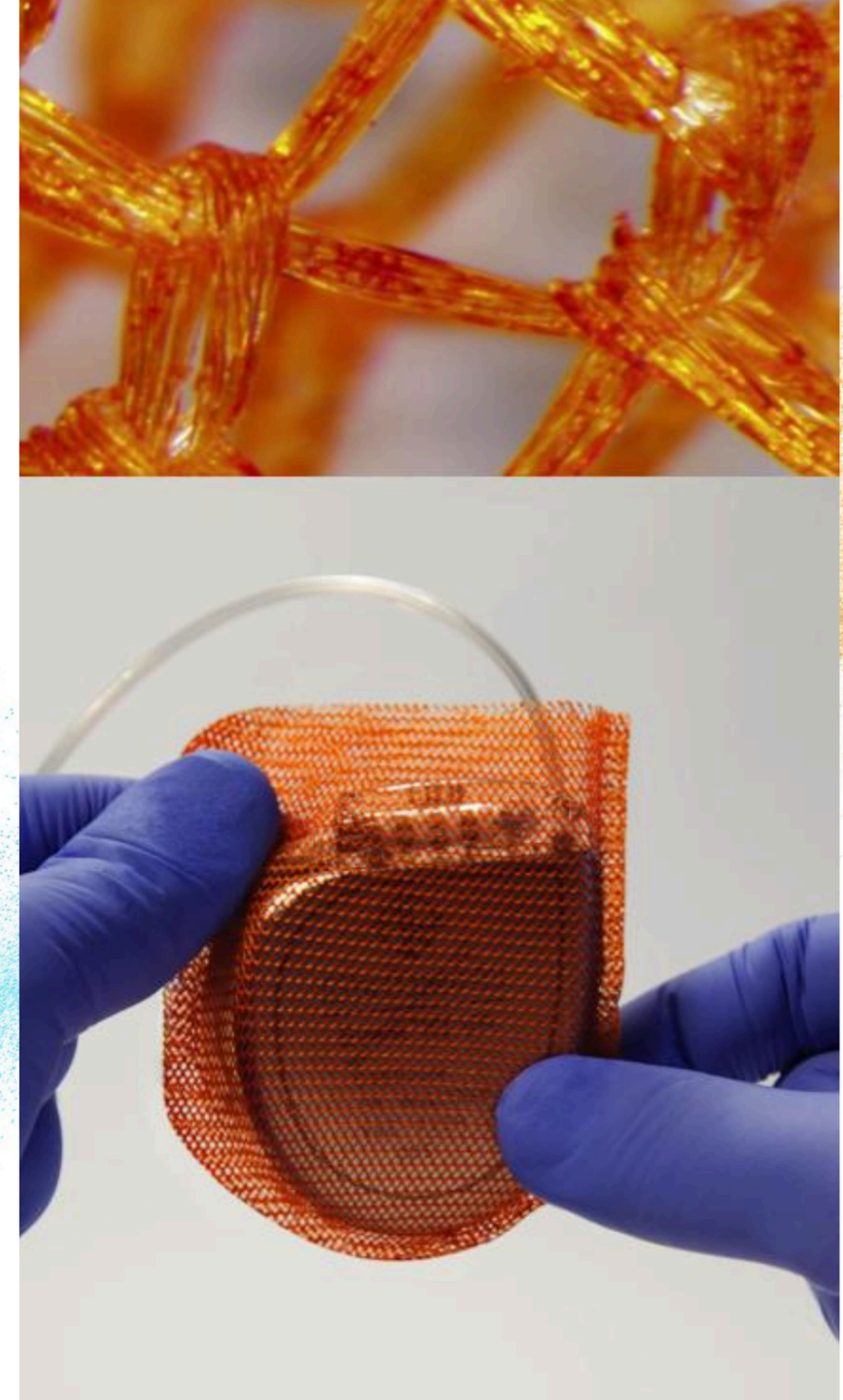
- 劑量只有全身給藥的約 **10%**，卻能在囊袋組織達到超過 MIC 的濃度
- 臨床前活體溶出研究：植入 2 小時內超過 MIC，至少維持 7 天
- 全身暴露極低，理論上藥物交互作用與全身副作用也低

全身劑量：Sanford Guide。包膜含量：TYRX 產品頁 (Medtronic)。

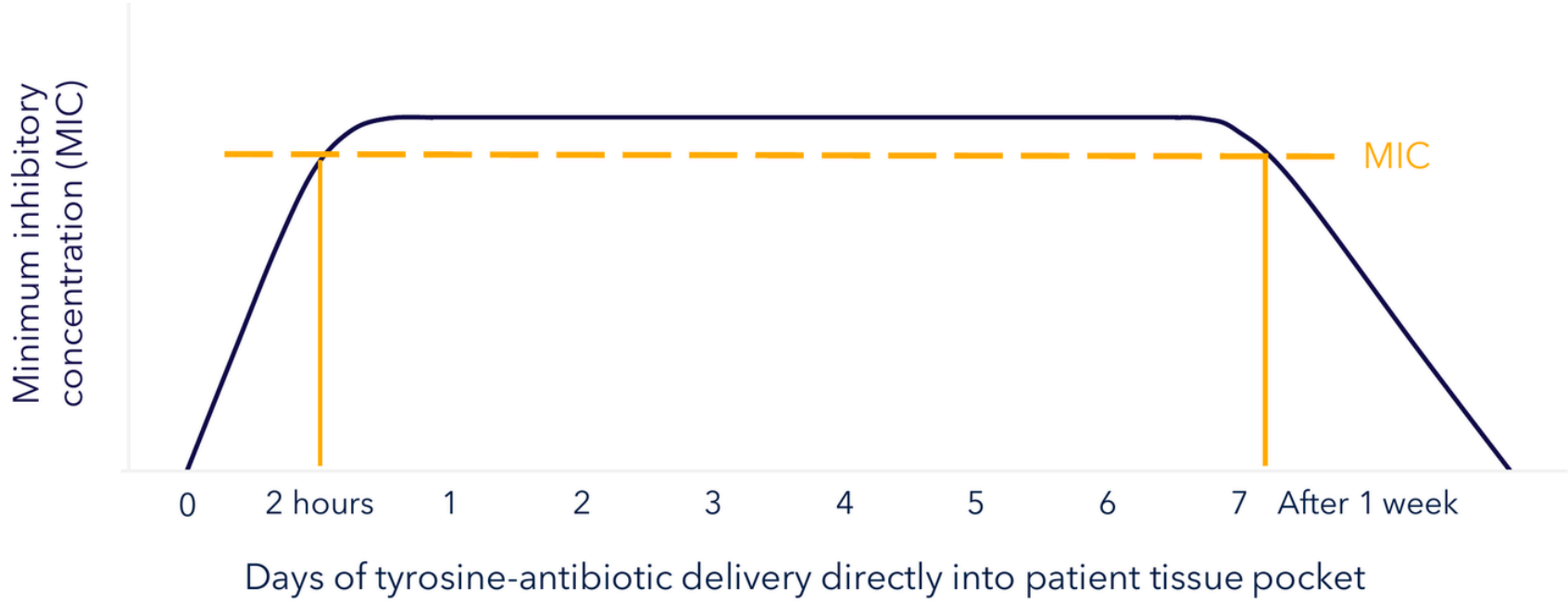
TYRX 可吸收抗菌包膜：一頁總結

- 經證實的 **CIED 穩定**作用
- 局部釋放的輔助性抗菌保護，**降低 CIED 感染**
- 兩種抗生素在囊袋組織內溶出，**2 小時**達 MIC，維持至少 **7 天**
- 多股網布約 **9 週**吸收；更換或修正手術時，手術技巧不需改變
- 吸收後**不留異物**，不會成為日後感染的病灶 (nidus)

感染降低的證據：WRAP-IT, NEJM 2019。穩定作用：Osoro M, et al. PACE 2018。



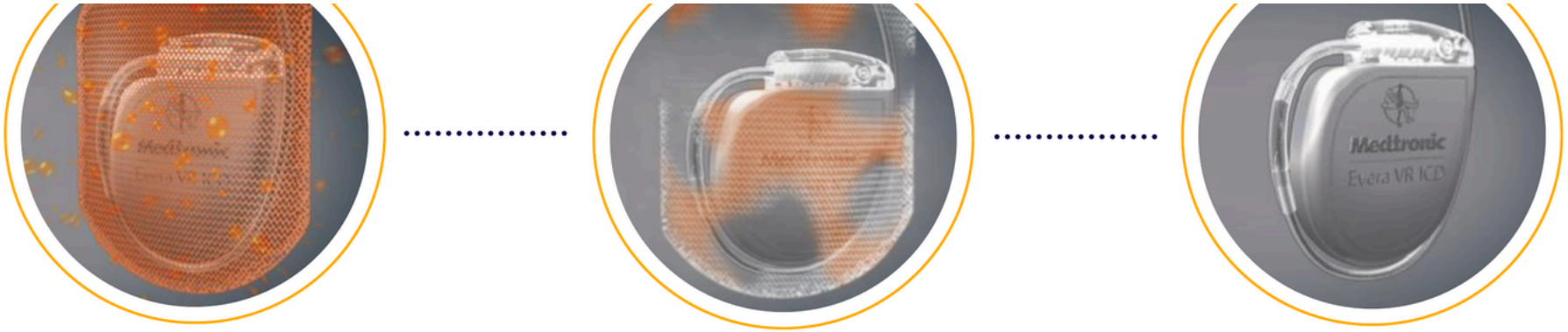
Tyrosine：「聚合物才是關鍵科學」



- Tyrosine 聚合物控制藥物釋放：**2 小時**內達 MIC，持續至少 **7 天**，之後逐漸下降
- 不是「一次倒進去」的 bolus，而是在最需要的時間窗維持有效濃度

示意圖取自原簡報；溶出動力學為 Medtronic data on file。動物感染模型中的實際效果：Hansen LK, et al. PACE 2009。

溶出與吸收的時間序列



Envelope after implantation[1]

Absorbable envelope is eluting minocycline and rifampicin

Envelope at four weeks[2]

Absorbable envelope is dissolving into fragments

Envelope at ~9 weeks[3]

Mesh has no visual presence and is fully absorbed

植入後	4 週	約 9 週
包膜持續釋放 minocycline 與 rifampin	包膜溶解成碎片	網布已無肉眼可見的殘留，完全吸收

Medtronic 臨床前資料，無公開連結。羊隻模型的吸收過程見 Virmani R, et al. Heart Rhythm 2024。

四、禁忌症與注意事項

仿單 (IFU) 規範

禁忌症：下列情況不應使用

- **全身性紅斑狼瘡 (SLE)**：minocycline 曾被報告會使病情惡化
- 對下列物質過敏或有過敏史：
  - Tetracyclines (minocycline)
  - Rifampin
  - 可吸收網布材質 (glycoprene II)
- **已污染或已感染的傷口**

依據 TYRX 仿單；Medtronic 產品頁。

注意事項

- 只有具備植入 pulse generator 或 defibrillator 資格的醫師才可使用
- Rifampin 與 minocycline 之間沒有已知的交互作用
- 與許多藥物一樣，minocycline 與 rifampin 若直接接觸**含碘溶液**，效果可能降低
- **不要改變**原本術前、術中、術後局部或全身抗生素的常規做法

> 一句話：TYRX 是輔助 (adjunctive)，不是取代。該給的靜脈抗生素還是要給。

依據 TYRX 仿單；Medtronic 產品頁。

對抗生素的過敏反應

植入後胸前瀰漫性紅疹。要同時考慮 minocycline、rifampin 與網布材質三種可能。台灣資料亦有 2 例包膜過敏：Chang CF, et al. Acta Cardiol Sin 2025。

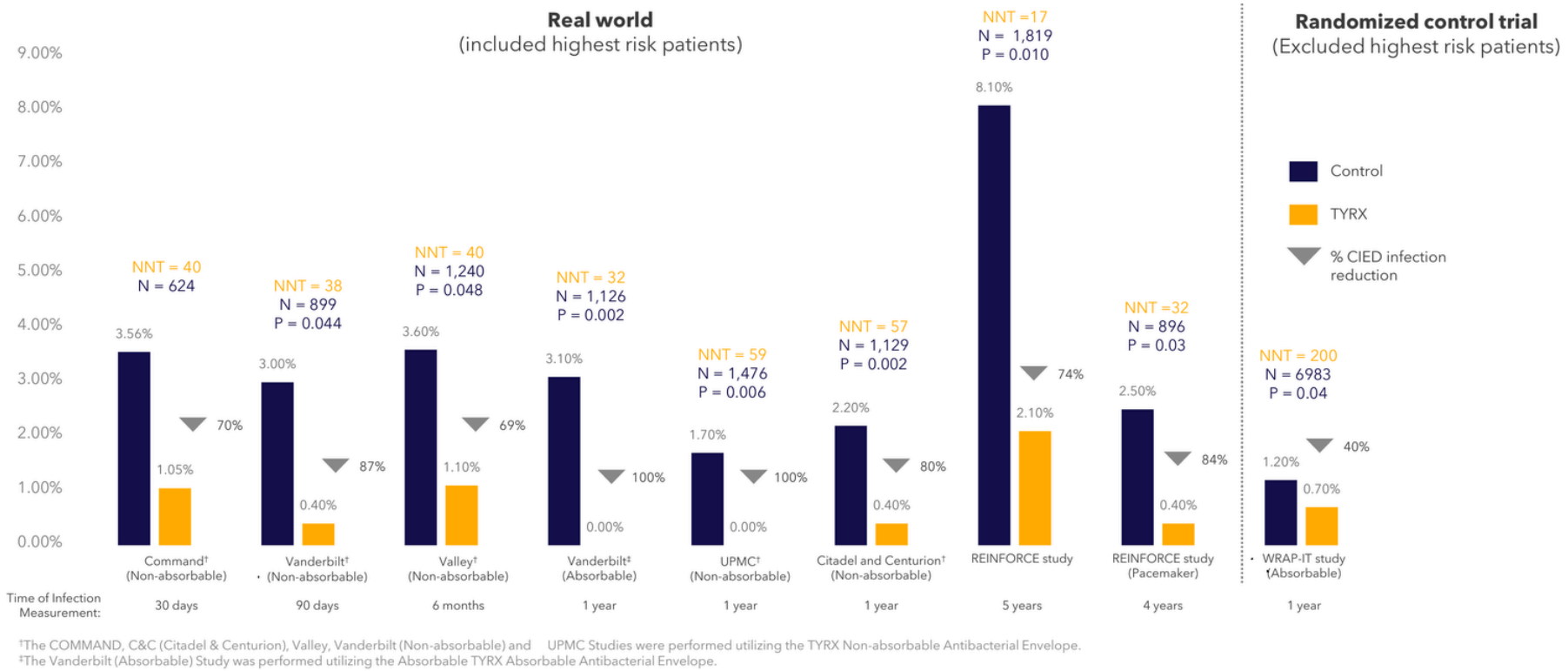
對網布材質的過敏反應

左：切口周圍大範圍紅斑。右：切口紅腫與滲出。需與早期囊袋感染鑑別。照片為原簡報病例，未發表。

五、臨床證據

WRAP-IT 隨機對照試驗

CIED 感染顯著降低：TYRX 臨床證據總覽



†The COMMAND, C&C (Citadel & Centurion), Valley, Vanderbilt (Non-absorbable) and UPMC Studies were performed utilizing the TYRX Non-absorbable Antibacterial Envelope.
‡The Vanderbilt (Absorbable) Study was performed utilizing the Absorbable TYRX Absorbable Antibacterial Envelope.

左：真實世界 (含最高風險病人)；右：WRAP-IT RCT。深色對照、淺色 TYRX。COMMAND (Henrikson 2017)、Vanderbilt (Kolek 2013)、Valley (Mittal 2014)、Vanderbilt abs. (Kolek 2015)、UPMC (Shariff 2015)、Citadel/Centurion (Bloom 2011)、REINFORCE (Ziacchi 2023)、REINFORCE PM (2025)、WRAP-IT (Tarakji 2019)。

WRAP-IT：全球最大 CIED RCT

6,983
高囊袋感染風險病人

776
植入醫師

181
中心

25
國家

納入條件：CIED 修正、電池更換、升級，或 de novo CRT-D。

Tarakji KG, Mittal S, Kennergren C, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：研究目的

> 評估在標準感染預防措施之外，加用 TYRX 包膜降低 CIED 感染的**安全性**與**有效性**。

試驗定義的 CIED 感染

- 表淺蜂窩性組織炎合併傷口裂開、磨穿或膿性分泌物，或
- 深部切口或 generator 囊袋感染，或
- 持續性菌血症，或
- 心內膜炎

Tarakji KG, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：主要感染 (major infection) 的定義

符合下列一項以上，即定義為主要 CIED 感染：

- CIED 系統**移除**
- 任何**侵入性處置**（例如打開囊袋）但未移除系統
- 病人不適合移除系統時，給予**延長抗生素治療**
- **死亡**

主要終點：12 個月內需系統拔除或修正、長期抗生素治療且感染復發，或死亡。Tarakji KG, et al. NEJM 2019。

WRAP-IT：試驗設計

- 前瞻性、隨機、對照、多中心、全球試驗
- 1:1 隨機分派至 TYRX 包膜組與對照組（不用 TYRX）
- 獨立臨床事件委員會：心臟電生理與感染科專家
- 獨立資料監測委員會
- 結果由 Cleveland Clinic Coordinating Center for Clinical Research 獨立驗證

納入：囊袋感染風險增加者

- 任何 CIED generator 更換、系統升級或修正
- 初次 CRT-D 植入

WRAP-IT：排除最高風險的病人

排除全身性感染風險最高的族群：

- de novo pacemaker 與 de novo ICD
- 血液透析或腹膜透析
- 免疫抑制劑（長期口服，或 prednisone ≥ 20 mg）
- 近 12 個月內或現存的感染

總結一句：WRAP-IT 證明的是中高風險族群的效益；最高風險族群要看真實世界資料。

Tarakji KG, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：主要結果

40%

主要 CIED 感染降低 (主要終點)

61%

囊袋 CIED 感染降低

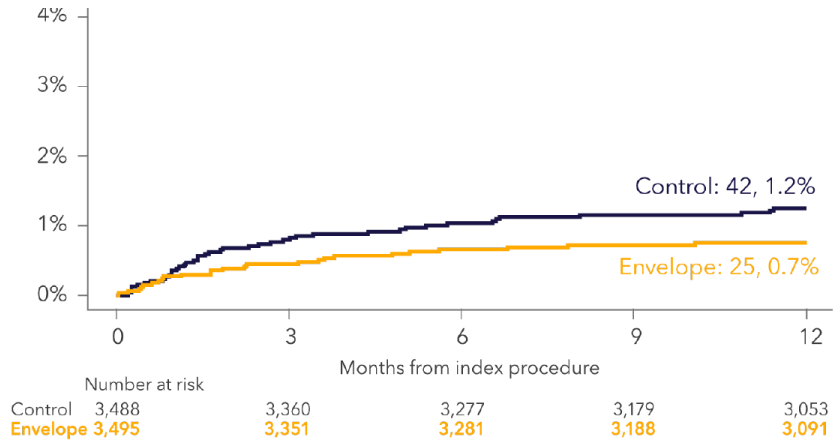
安全達標

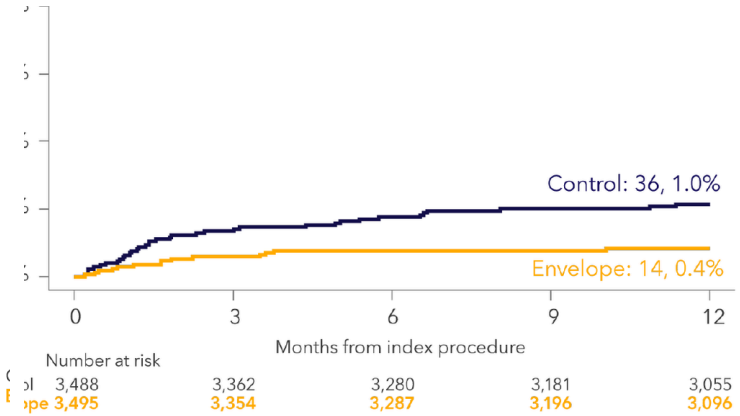
12 個月內未增加併發症風險

這些結果為 TYRX 包膜提供了目前最完整的 CIED 感染預防證據。

Tarakji KG, Mittal S, Kennergren C, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：Kaplan-Meier 曲線





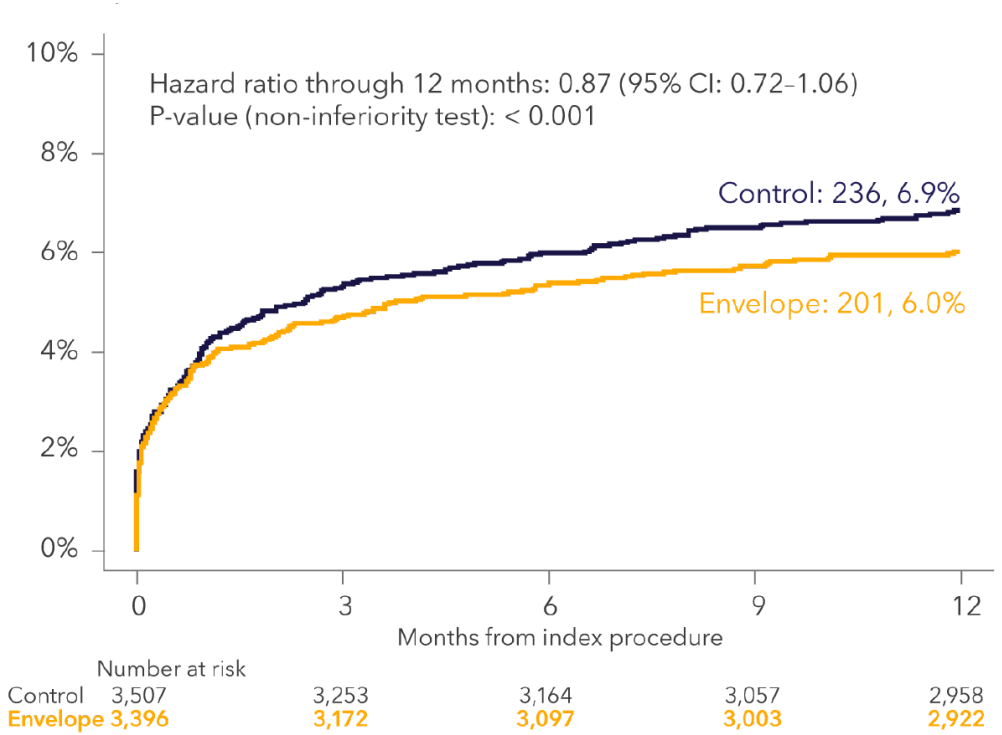
75% of initial major CIED infecti vere pocket infections.

- **左、主要感染**：HR 0.60 (95% CI 0.36-0.98)，p = 0.04。12 個月 0.7% vs 1.2%
- **右、囊袋感染**：HR 0.39 (95% CI 0.21-0.72)，p < 0.01。0.4% vs 1.0%
- 初次主要 CIED 感染中，**75%** 是囊袋感染

Tarakji KG, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：併發症未增加

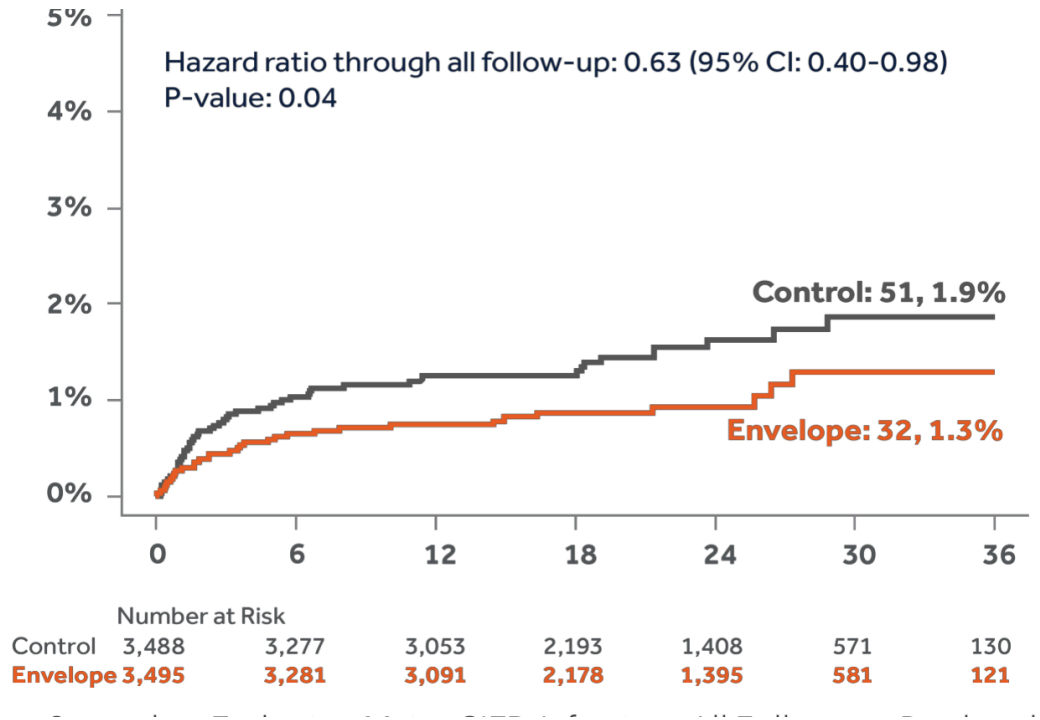
- 預先設定的次要分析 (as-treated, 非劣性)
- 12 個月併發症：包膜組 **6.0%** vs 對照組 **6.9%**
- HR 0.87 (95% CI 0.72-1.06)，非劣性 p < 0.001
- 排除主要感染後，12 個月 KM 併發症率 5.7% vs 5.9%



Tarakji KG, et al. N Engl J Med 2019;380:1895-1905。

WRAP-IT：長期追蹤效果持續

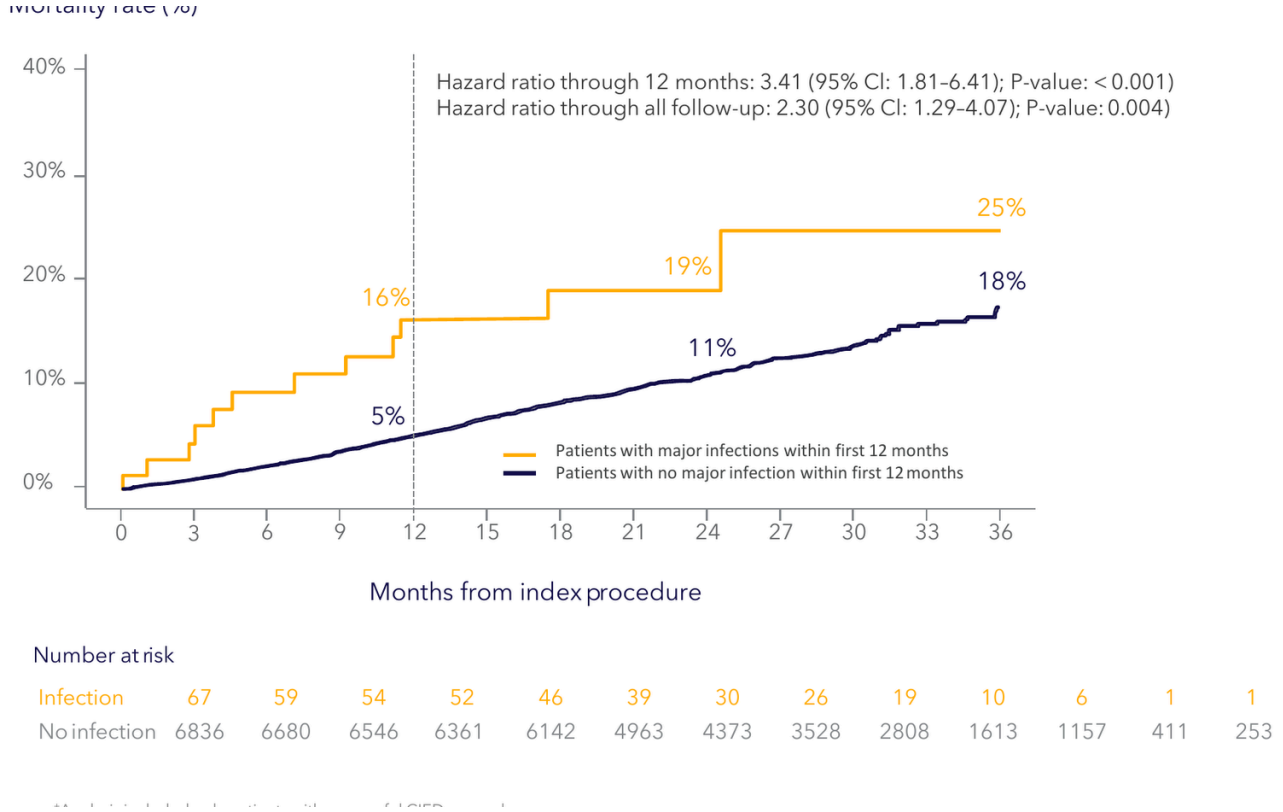
- 次要終點：全追蹤期間的主要 CIED 感染
- 平均追蹤 20.7 ± 8.5 個月
- 包膜組 32 例 (1.3%) vs 對照組 51 例 (1.9%)
- HR 0.63 (95% CI 0.40-0.98)，p = 0.04
- 注意：此 p 值未經多重比較校正；預先設定的校正後，**未達顯著**



Tarakji KG, et al. N Engl J Med 2019;380:1895-1905。

主要 CIED 感染後，一年死亡風險超過 3 倍

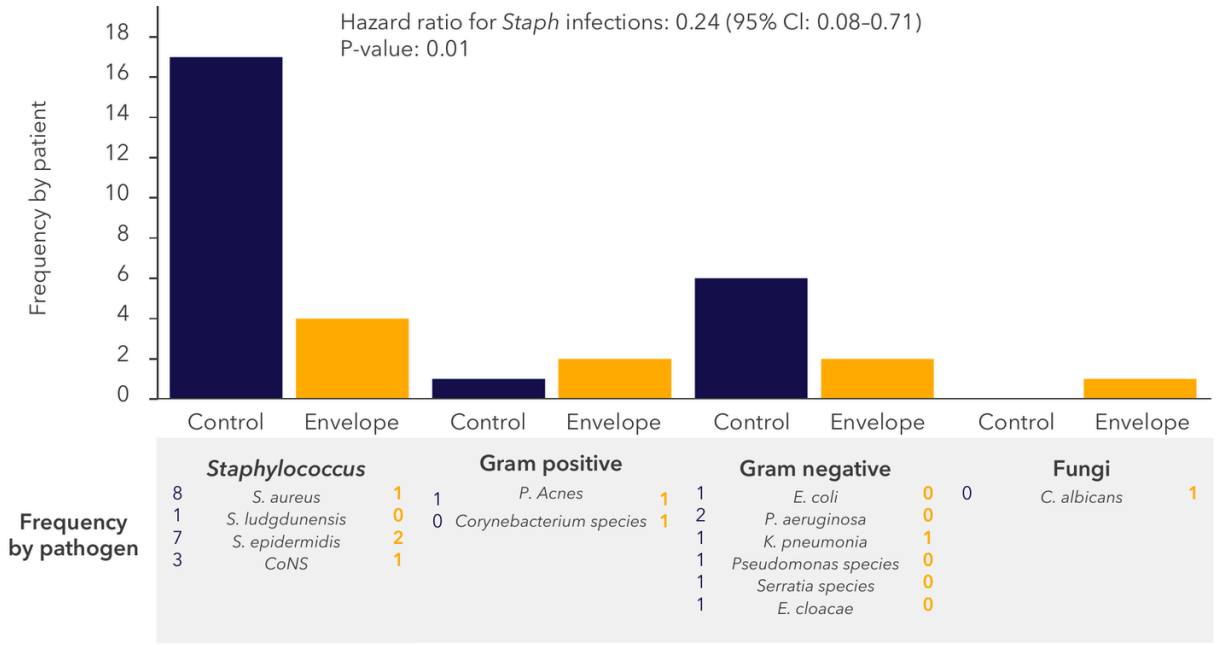
- WRAP-IT 預先設定分析，n = 6,903
- 12 個月：HR **3.41** (95% CI 1.81-6.41)，p < 0.001
- 全追蹤期：HR **2.30** (95% CI 1.29-4.07)，p = 0.004
- 感染者生活品質下降達 6 個月；36% 治療中斷，中位 184 天



Wilkoff BL, Boriani G, Mittal S, et al. Circ Arrhythm Electrophysiol 2020;13:e008280。

對 Staph 囊袋感染的顯著效果：微生物學分析

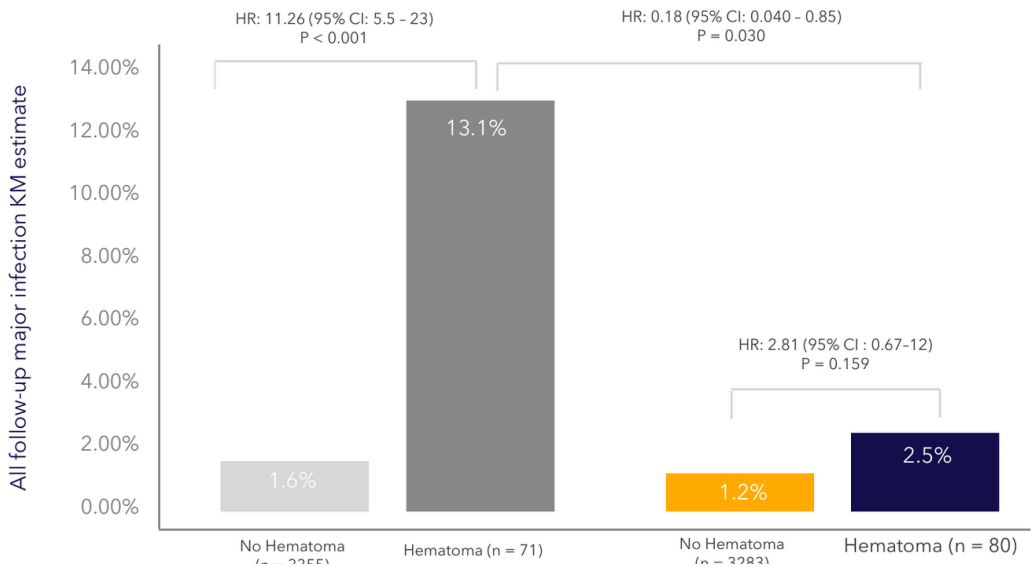
- Per-protocol 分析，n = 6,800
- **Staphylococcus 感染 HR 0.24** (95% CI 0.08-0.71)，p = 0.01
- 對照組 17 例 vs 包膜組 4 例
- Gram (+) 其他菌、gram (-) 與黴菌的例數都少，看不出差異



Sohail MR, Corey GR, Wilkoff BL, et al. IDWeek 2019 摘要 849. Open Forum Infect Dis 2019;6(Suppl 2)。

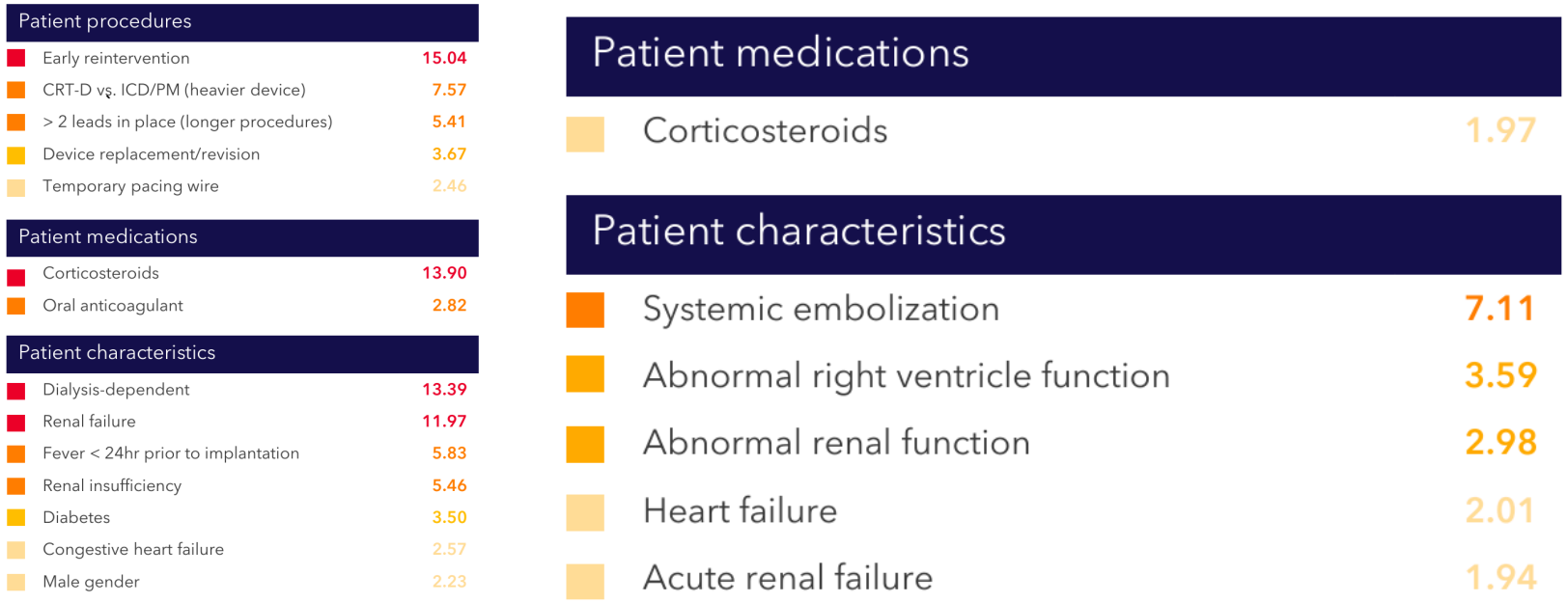
血腫 (hematoma) 時，TYRX 降低 82% 囊袋感染

- 30 天血腫率 2.22%，兩組無差異 (p = 0.39)
- **對照組**：有血腫者感染 13.1% vs 無血腫 1.6%，HR 11.26，p < 0.001
- **包膜組**：有血腫者 2.5% vs 無血腫 1.2%，HR 2.81，p = 0.159
- 血腫病人：包膜 vs 對照 HR **0.18** (95% CI 0.04-0.85)，p = 0.030
- 血腫者的主要感染全是囊袋感染，沒有菌血症或心內膜炎



Tarakji KG, et al. Heart Rhythm 2021;18:2080-2086。

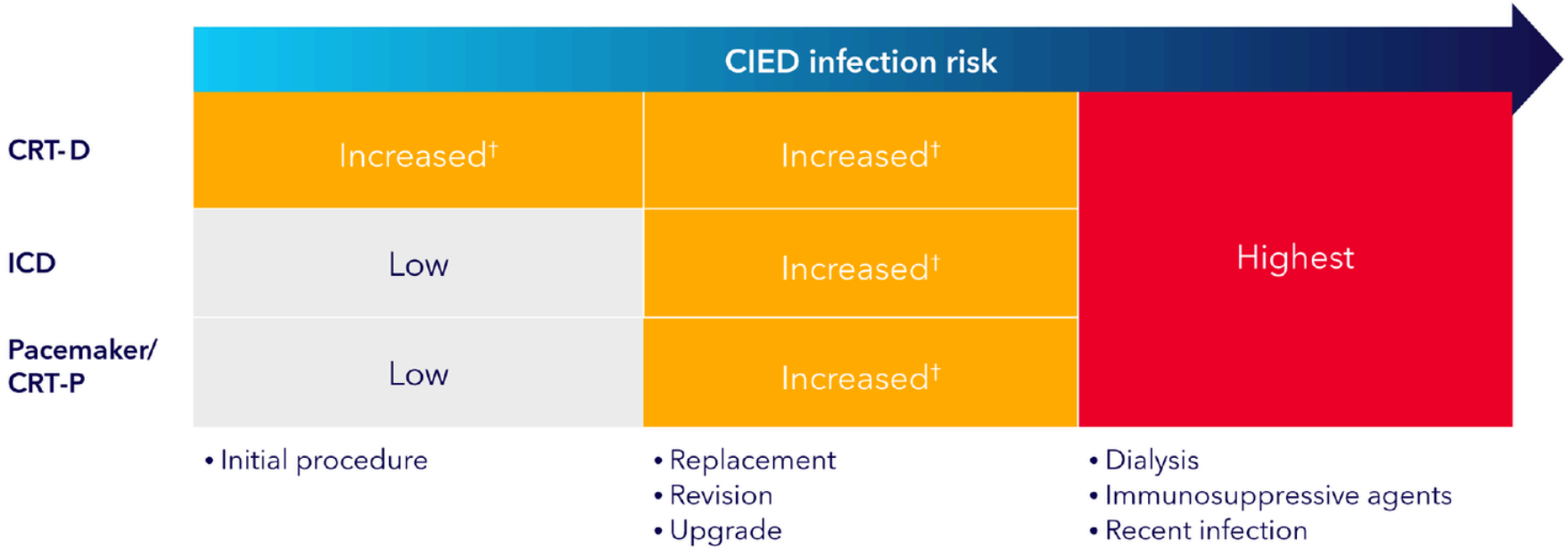
CIED 感染的風險因子與死亡率



- **左、感染的 odds ratio**：早期再介入 15.0、類固醇 13.9、透析 13.4、腎衰竭 12.0
- **右、感染後死亡的 hazard ratio**：全身性栓塞 7.1、右心室功能異常 3.6

Klug 2007；Lekkerkerker 2009；Margey 2010；Sohail 2007；Bloom 2006；Hercé 2013；Romeyer-Bouchard 2010；Habib 2013；Baman 2009。

風險因子與病人選擇



- **風險增加 (RCT 資料)**：12 個月主要感染 1.0%；TYRX 降低 40% 主要感染、61% 囊袋感染
- **最高風險 (真實世界)**：12 個月主要感染 1 到 4%；TYRX 降低 70 到 100% 主要感染
- 選擇病人時，也要把「固定裝置、提供穩定環境」的效益考慮進去

Tarakji 2019；Bloom 2011；Mittal 2014；Kolek 2013；Kolek 2015；Shariff 2015；Henrikson 2017；另引 Medtronic data on file。

六、真實世界資料

REINFORCE、台灣經驗、指引與共識

REINFORCE：義大利真實世界

- 24 個義大利中心，1,819 位連續病人
- TYRX 872 人 (47.9%)、對照 947 人
- PADIT ≥ 7：TYRX 組 37.3%，對照組 11.4%
- 平均追蹤 1.4 年：感染相關事件 **0.8% vs 2.4%**，p = 0.007
- 5 年累積發生率 **2.1% vs 8.1%**，HR 0.34，p = 0.010
- **NNT = 17**

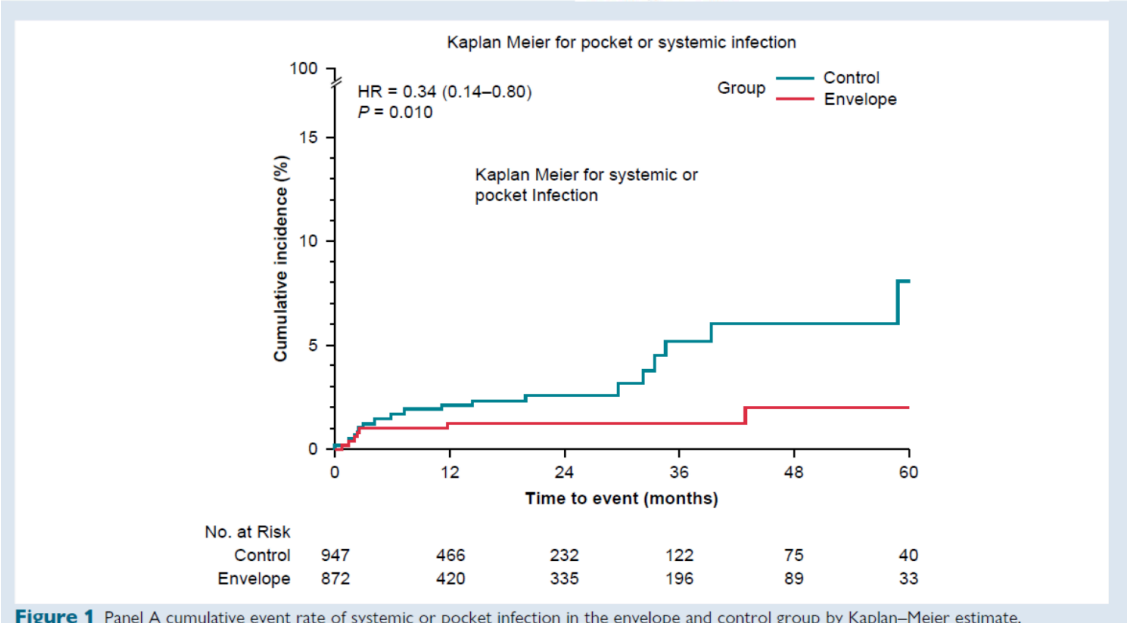


Figure 1 Panel A cumulative event rate of systemic or pocket infection in the envelope and control group by Kaplan–Meier estimate.

Ziacchi M, Biffi M, Iacopino S, et al. Europace 2023;25:euad224。

REINFORCE：各風險層都有效

PADIT 分層	TYRX	對照	p
低	0.0%	1.7%	0.029
中	0.7%	2.9%	0.064
高	1.5%	5.6%	0.022

> Suneet Mittal：「低風險組的感染率仍有臨床意義，而包膜在所有風險組都降低感染。看來不能只靠風險分數決定不用包膜。」

Ziacchi M, et al. Europace 2023。

Baseline characteristic	Total (n = 1,819)	Envelope (n = 872)	Control (n = 947)	p-value
De Novo CIED	64.7% (1178)	45% (393)	82.8% (785)	< 0.001
De Novo PM	52.9% (623)	51.2% (201)	53.8% (422)	0.457
De Novo ICD	47.1% (555)	48.8% (192)	46.2% (363)	0.457

At least on infection-related clinical event (pocket or systematic infection)	Total (n = 1,819)	Envelope (n = 872)	Control (n = 947)	p-value
PADIT Score = Low	1.2% (11/903)	0.0% (0/271)	1.7% (11/632)	0.029
PADIT Score = Medium	1.7% (8/483)	0.7% (2/276)	2.9% (6/207)	0.064
PADIT Score = High	2.5% (11/433)	1.5% (5/325)	5.6% (6/108)	0.022

REINFORCE：追蹤 5 年的額外觀察

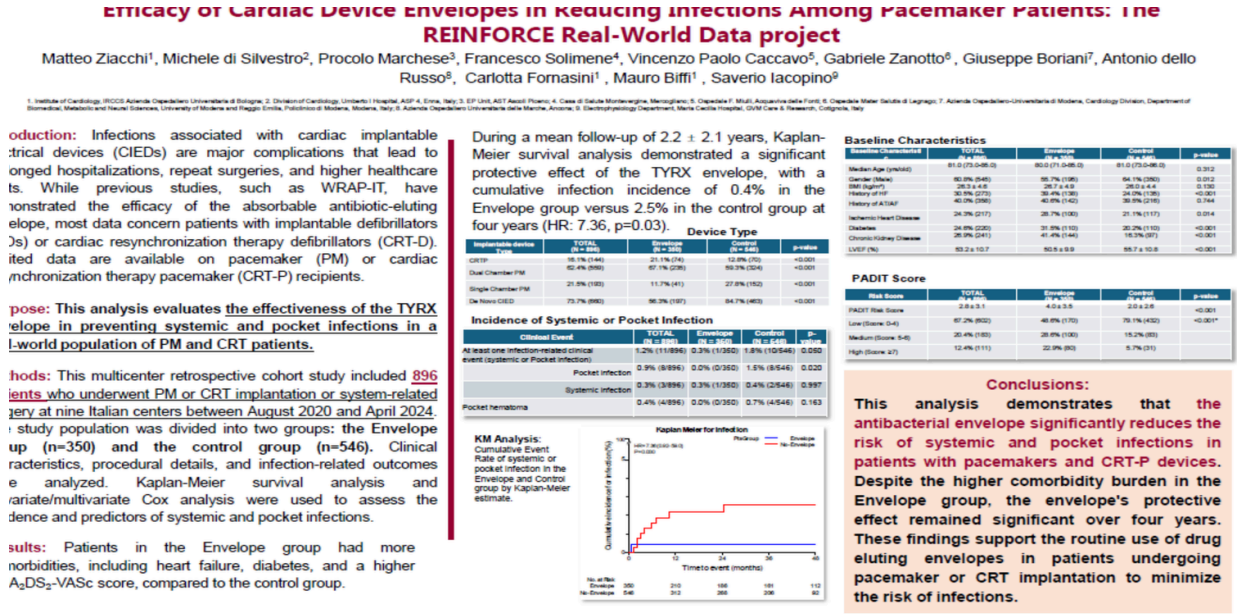
- 對照組的感染率偏高：1.4 年 2.4%，5 年 8.1%
- De novo CIED：TYRX 0.8% vs 對照 1.9%，p = 0.130，趨勢向好但未達顯著
- 不分病人風險類型，1.4 年時 TYRX 都與較低感染相關
- 傾向分數配對 (propensity score matching) 585 對，PADIT 分數平衡後結論相同

> 包膜在高、中、低風險族群都降低逾 60% 的感染相關事件，效果長期維持。

Ziacchi M, et al. Europace 2023。成本效益：Wilkoff BL, et al. Circ AE 2020；Boriani G, et al. Value Health 2021;24:930-938。

REINFORCE Brady：只看 pacemaker 與 CRT-P

- 多中心回溯分析，2020-08 至 2024-04
- TYRX 組 350 人 vs 對照組 546 人，中位追蹤 2.2 年
- PADIT ≥ 7：TYRX 組 22.9% vs 對照組 5.7%
- 感染發生 0.3% vs 1.8%；incidence rate ratio 0.11



Efficacy of Cardiac Device Envelopes in Reducing Infections Among Pacemaker Patients: The REINFORCE Real-World Data project

Matteo Ziacchi[1], Michele di Silvestro[2], Procolo Marchese[3], Francesco Solimene[4], Vincenzo Paolo Caccavo[5], Gabriele Zanotto[6], Giuseppe Boriani[7], Antonio dello Russo[8], Carlotta Fornasini[1], Mauro Biffi[1], Saverio Iacopino[9]

oduction: Infections associated with cardiac implantable trical devices (CIEDs) are major complications that lead to onged hospitalizations, repeat surgeries, and higher healthcare ts. While previous studies, such as WRAP-IT, have nonstrated the efficacy of the absorbable antibiotic-eluting elope, most data concern patients with implantable defibrillators Ds) or cardiac resynchronization therapy defibrillators (CRT-D). ited data are available on pacemaker (PM) or cardiac nchronization therapy pacemaker (CRT-P) recipients.

pose: This analysis evaluates the effectiveness of the TYRX elope in preventing systemic and pocket infections in a l-world population of PM and CRT patients.

hods: This multicenter retrospective cohort study included 896 ients who underwent PM or CRT implantation or system-related gery at nine Italian centers between August 2020 and April 2024. study population was divided into two groups: the Envelope up (n=350) and the control group (n=546). Clinical racteristics, procedural details, and infection-related outcomes e analyzed. Kaplan-Meier survival analysis and ariate/multivariate Cox analysis were used to assess the dence and predictors of systemic and pocket infections.

sults: Patients in the Envelope group had more norbidities, including heart failure, diabetes, and a higher A_2DS_2-VASc score, compared to the control group.

During a mean follow-up of 2.2 ± 2.1 years, Kaplan-Meier survival analysis demonstrated a significant protective effect of the TYRX envelope, with a cumulative infection incidence of 0.4% in the Envelope group versus 2.5% in the control group at four years (HR: 7.36, p=0.03).

Device Type

Implantable device Type	TOTAL (N = 896)	Envelope (N = 350)	Control (N = 546)	p-value
CRTP	16.1% (144)	21.1% (74)	12.8% (70)	<0.001
Dual Chamber PM	62.4% (559)	67.1% (235)	59.3% (324)	<0.001
Single Chamber PM	21.5% (193)	11.7% (41)	27.8% (152)	<0.001
De Novo CIED	73.7% (660)	66.3% (197)	84.7% (463)	<0.001

Incidence of Systemic or Pocket Infection

Clinical Event	TOTAL (N = 896)	Envelope (N = 350)	Control (N = 546)	p-value
At least one infection-related clinical event (systemic or Pocket infection)	1.2% (11/896)	0.3% (1/350)	1.8% (10/546)	0.050
Pocket infection	0.9% (8/896)	0.0% (0/350)	1.5% (8/546)	0.020
Systemic infection	0.3% (3/896)	0.3% (1/350)	0.4% (2/546)	0.997
Pocket hematoma	0.4% (4/896)	0.0% (0/350)	0.7% (4/546)	0.163

KM Analysis: Cumulative Event Rate of systemic or pocket infection in the Envelope and Control group by Kaplan-Meier estimate.

Baseline Characteristics

Baseline Characteristics	TOTAL (N = 896)	Envelope (N = 350)	Control (N = 546)	p-value
Median Age (yrs[IQR])	81.0 (73.0-86.0)	80.0 (71.0-85.0)	81.0 (73.0-86.0)	0.312
Gender (Male)	60.8% (545)	58.7% (198)	64.1% (350)	0.012
BMI (kg/m²)	26.3 ± 4.6	26.7 ± 4.9	26.0 ± 4.4	0.130
History of HF	30.5% (273)	39.4% (138)	24.0% (135)	<0.001
History of AFIAF	40.0% (356)	40.8% (142)	39.5% (216)	0.744
Ischemic Heart Disease	24.3% (217)	28.7% (100)	21.1% (117)	0.014
Diabetes	24.6% (220)	31.5% (110)	20.2% (110)	<0.001
Chronic Kidney Disease	26.9% (241)	41.4% (144)	15.3% (97)	<0.001
LVEF (%)	53.2 ± 10.7	50.5 ± 9.9	55.7 ± 10.6	<0.001

PADIT Score

Risk Score	TOTAL (N = 896)	Envelope (N = 350)	Control (N = 546)	p-value
PADIT Risk Score	2.8 ± 3.1	4.0 ± 3.5	2.0 ± 2.6	<0.001
Low (Score: 0-4)	67.2% (602)	48.6% (170)	79.1% (432)	<0.001*
Medium (Score: 5-6)	20.4% (183)	28.6% (100)	15.2% (83)	
High (Score: ≥7)	12.4% (111)	22.9% (80)	5.7% (31)	

Conclusions:

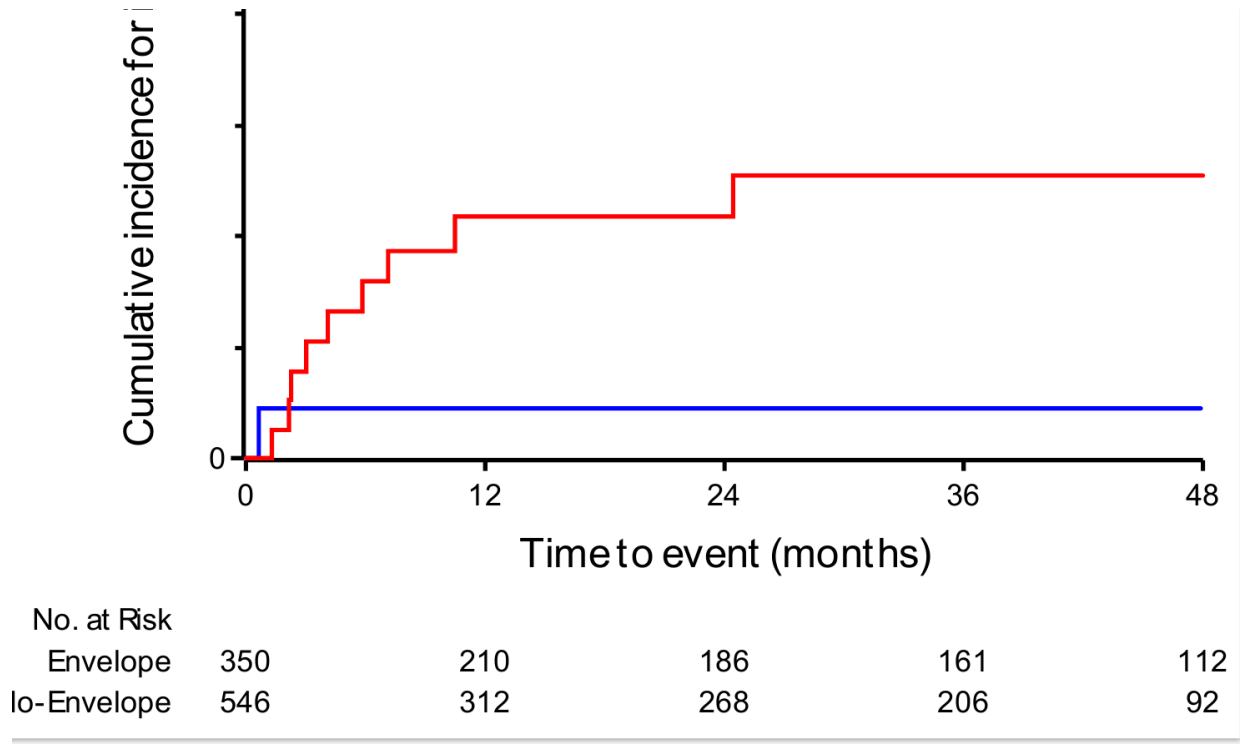
This analysis demonstrates that the antibacterial envelope significantly reduces the risk of systemic and pocket infections in patients with pacemakers and CRT-P devices. Despite the higher comorbidity burden in the Envelope group, the envelope's protective effect remained significant over four years. These findings support the routine use of drug eluting envelopes in patients undergoing pacemaker or CRT implantation to minimize the risk of infections.

Ziacchi M, di Silvestro M, Marchese P, et al. Heart Rhythm O2 2025;6:1356-1363。

REINFORCE Brady：4 年 Kaplan-Meier

- 4 年累積感染：TYRX **0.4%** vs 對照 **2.5%**
- HR 7.36 (95% CI 0.92-59.0)，p = 0.03
- **NNT = 32**
- 比較特別的是，這群人過去被認為風險較低，包膜仍有效果

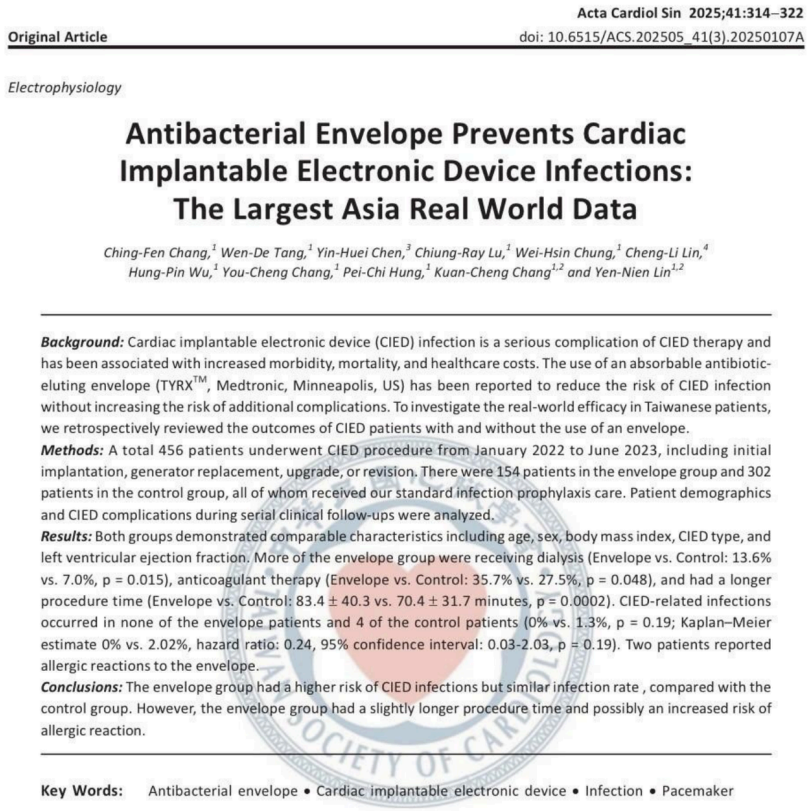
EHRA 2025 發表；正式論文 Ziacchi M, et al. Heart Rhythm O2 2025。



台灣資料：中國醫藥大學附設醫院

- 回溯分析，2022-01 至 2023-06，456 位病人
- 包膜組 154 人、對照組 302 人，皆接受標準預防
- 包膜組透析比例較高 (13.6% vs 7.0%)、抗凝比例較高、手術時間較長 (83 vs 70 分鐘)
- CIED 感染：**0 例 vs 4 例** (0% vs 1.3%，p = 0.19)
- KM 估計 0% vs 2.02%，HR 0.24 (95% CI 0.03-2.03)，**未達統計顯著**
- 2 位病人出現包膜過敏反應

Chang CF, Tang WD, Chen YH, et al. Acta Cardiol Sin 2025;41:314-322 (DOI 直接開啟全文 PDF)；PMC 全文。



Acta Cardiol Sin 2025;41:314–322
Original Article
doi: 10.6515/ACS.202505_41(3).20250107A
Electrophysiology

Antibacterial Envelope Prevents Cardiac Implantable Electronic Device Infections: The Largest Asia Real World Data

Ching-Fen Chang,[1] Wen-De Tang,[1] Yin-Huei Chen,[3] Chiung-Ray Lu,[1] Wei-Hsin Chung,[1] Cheng-Li Lin,[4] Hung-Pin Wu,[1] You-Cheng Chang,[1] Pei-Chi Hung,[1] Kuan-Cheng Chang[1,2] and Yen-Nien Lin[1,2]

Background: Cardiac implantable electronic device (CIED) infection is a serious complication of CIED therapy and has been associated with increased morbidity, mortality, and healthcare costs. The use of an absorbable antibiotic-eluting envelope (TYRX™, Medtronic, Minneapolis, US) has been reported to reduce the risk of CIED infection without increasing the risk of additional complications. To investigate the real-world efficacy in Taiwanese patients, we retrospectively reviewed the outcomes of CIED patients with and without the use of an envelope.
Methods: A total 456 patients underwent CIED procedure from January 2022 to June 2023, including initial implantation, generator replacement, upgrade, or revision. There were 154 patients in the envelope group and 302 patients in the control group, all of whom received our standard infection prophylaxis care. Patient demographics and CIED complications during serial clinical follow-ups were analyzed.
Results: Both groups demonstrated comparable characteristics including age, sex, body mass index, CIED type, and left ventricular ejection fraction. More of the envelope group were receiving dialysis (Envelope vs. Control: 13.6% vs. 7.0%, p = 0.015), anticoagulant therapy (Envelope vs. Control: 35.7% vs. 27.5%, p = 0.048), and had a longer procedure time (Envelope vs. Control: 83.4 ± 40.3 vs. 70.4 ± 31.7 minutes, p = 0.0002). CIED-related infections occurred in none of the envelope patients and 4 of the control patients (0% vs. 1.3%, p = 0.19; Kaplan–Meier estimate 0% vs. 2.02%, hazard ratio: 0.24, 95% confidence interval: 0.03-2.03, p = 0.19). Two patients reported allergic reactions to the envelope.
Conclusions: The envelope group had a higher risk of CIED infections but similar infection rate , compared with the control group. However, the envelope group had a slightly longer procedure time and possibly an increased risk of allergic reaction.

Key Words: Antibacterial envelope • Cardiac implantable electronic device • Infection • Pacemaker

台灣資料：Kaplan-Meier

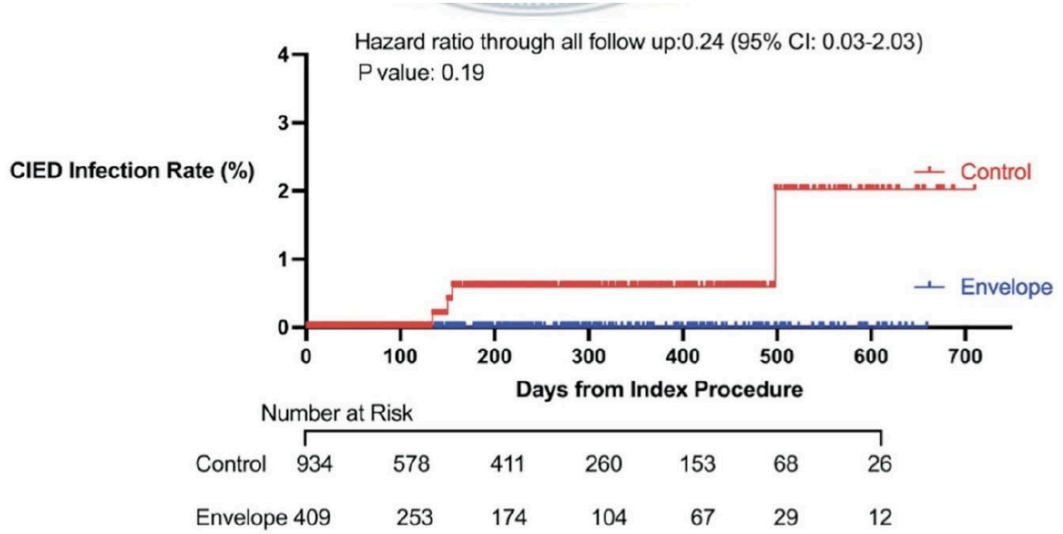


Figure 1. *Kaplan-Meier for systemic or pocket infections rate of cardiac implantable electronic device (CIED). CI, confidence interval.*

> 樣本數小、事件數少，是這篇無法達到顯著的主因。不過，風險較高的包膜組感染率為零，方向與國外資料一致。過敏反應則是亞洲資料中值得留意的訊號。

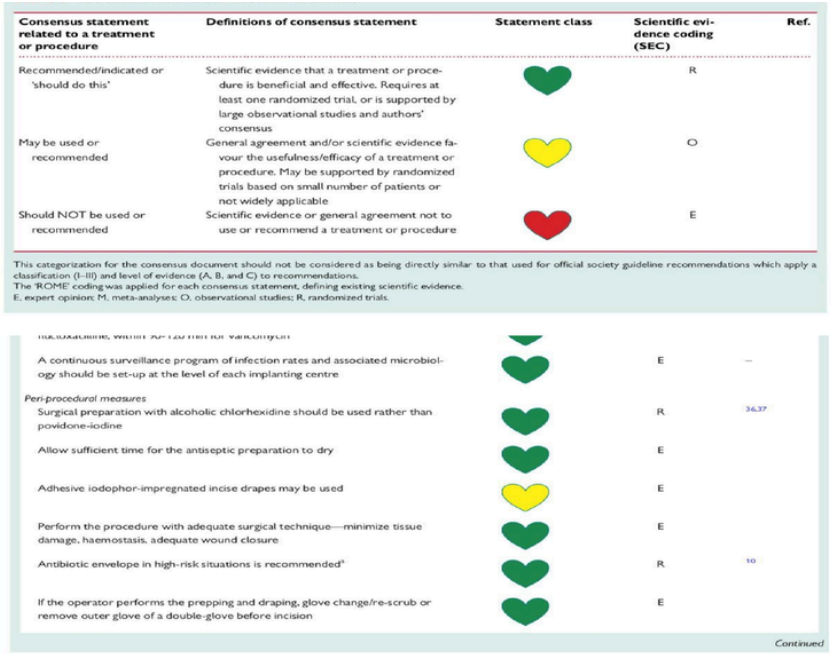
Chang CF, et al. Acta Cardiol Sin 2025;41:314-322。

ESC/EHRA 國際共識文件 (2020)

ESC European Society of Cardiology

EHRA CONSENSUS PAPER

European Heart Rhythm Association (EHRA) international consensus document on how to prevent, diagnose, and treat cardiac implantable electronic device infections—endorsed by the Heart Rhythm Society (HRS), the Asia Pacific Heart Rhythm Society (APHRS), the Latin American Heart Rhythm Society (LAHRS), International Society for Cardiovascular Infectious Diseases (ISCVID) and the European Society of Clinical Microbiology and Infectious Diseases (ESCMID) in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS)



Consensus statement related to a treatment or procedure	Definitions of consensus statement	Statement class	Scientific evidence coding (SEC)	Ref.
Recommended/indicated or 'should do this'	Scientific evidence that a treatment or procedure is beneficial and effective. Requires at least one randomized trial, or is supported by large observational studies and authors' consensus		R	
May be used or recommended	General agreement and/or scientific evidence favour the usefulness/efficacy of a treatment or procedure. May be supported by randomized trials based on small number of patients or not widely applicable		O	
Should NOT be used or recommended	Scientific evidence or general agreement not to use or recommend a treatment or procedure		E	

This categorization for the consensus document should not be considered as being directly similar to that used for official society guideline recommendations which apply a classification (I–III) and level of evidence (A, B, and C) to recommendations.
The 'ROME' coding was applied for each consensus statement, defining existing scientific evidence.
E, expert opinion; M, meta-analyses; O, observational studies; R, randomized trials.

A continuous surveillance program of infection rates and associated microbiology should be set-up at the level of each implanting centre		E	–
Peri-procedural measures			
Surgical preparation with alcoholic chlorhexidine should be used rather than povidone-iodine		R	34,37
Allow sufficient time for the antiseptic preparation to dry		E	
Adhesive iodophor-impregnated incise drapes may be used		E	
Perform the procedure with adequate surgical technique—minimize tissue damage, haemostasis, adequate wound closure		E	
Antibiotic envelope in high-risk situations is recommended[a]		R	10
If the operator performs the prepping and draping, glove change/re-scrub or remove outer glove of a double-glove before incision		E	

Continued

- 抗菌包膜在高風險病人**可考慮使用** (may be used or recommended)
- 同一文件亦涵蓋術前靜脈抗生素、皮膚準備與血腫的預防處理

Blomström-Lundqvist C, Traykov V, Erba PA, et al. Europace 2020;22:515-549。

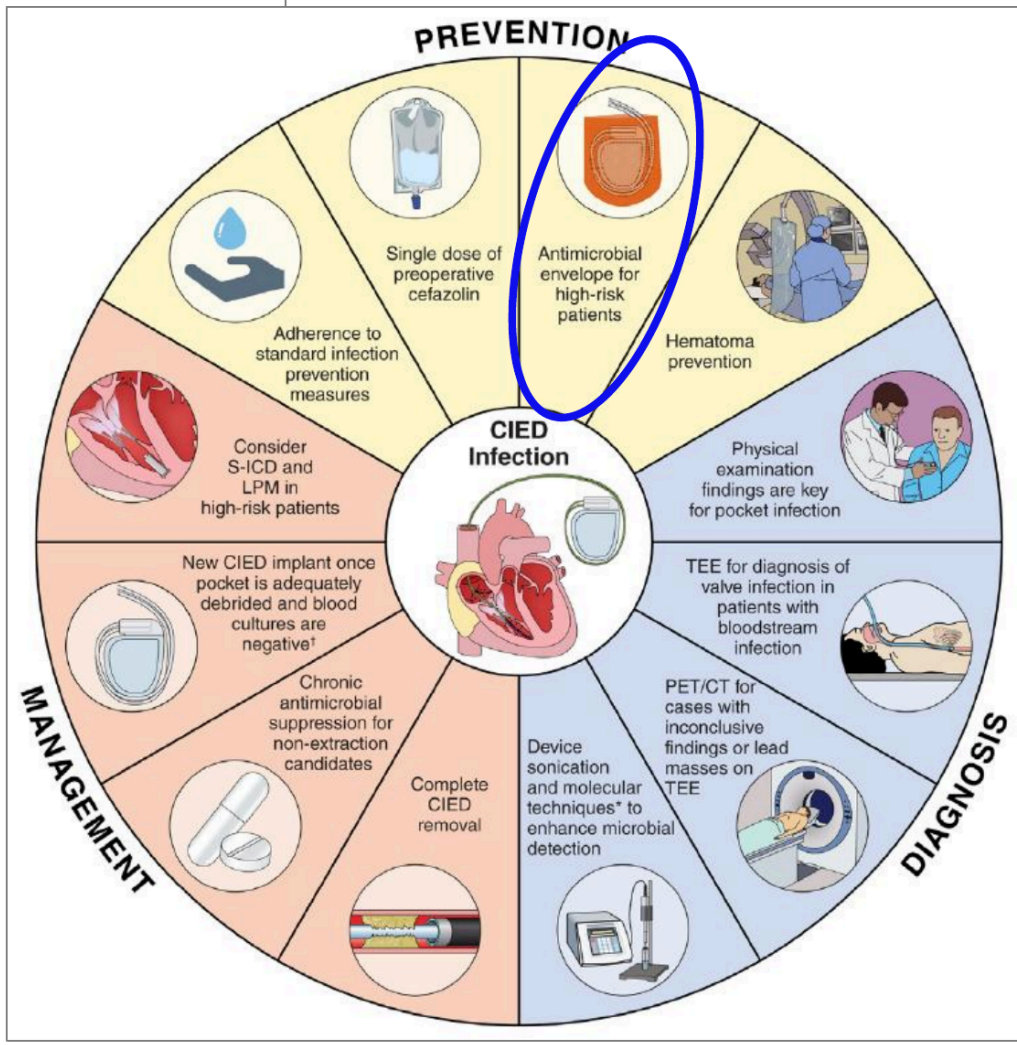
AHA 科學聲明 (2024)：CIED 感染的預防、診斷與處置

- 回顧 2010 年以來的證據與重大進展
- 確診 CIED 感染需**完整移除裝置**；早期移除與較好預後相關
- **血腫**是顯著且獨立的感染風險因子
- 高風險病人**可考慮抗菌包膜**
- 文件引用 WRAP-IT 的 61% 囊袋感染降低，以及血腫時 82% 的降低

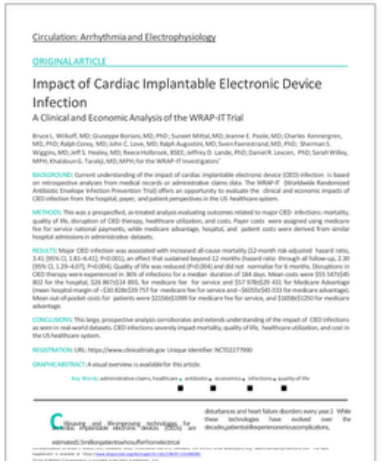
Baddour LM, Esquer Garrigos Z, Sohail MR, et al. Circulation 2024;149:e201-e216。

AHA SCIENTIFIC STATEMENT

Update on Cardiovascular Implantable E
Device Infections and Their Prevention,
and Management: A Scientific Statemer
the American Heart Association



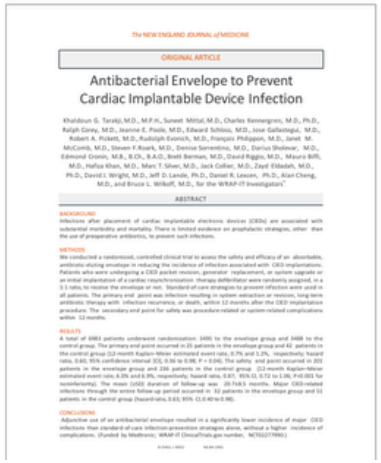
證據鏈已經完整


Circulation: Arrhythmia and Electrophysiology

ORIGINAL ARTICLE

Impact of Cardiac Implantable Electronic Device Infection

A Clinical and Economic Analysis of the WRAP-IT Trial


The NEW ENGLAND JOURNAL of MEDICINE

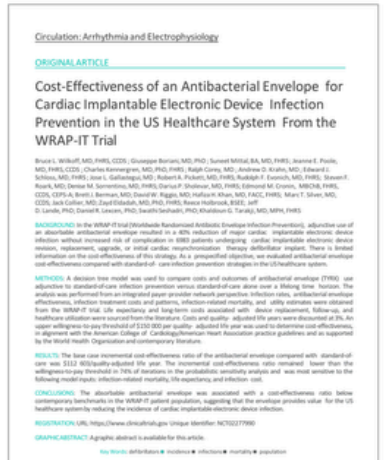
ORIGINAL ARTICLE

Antibacterial Envelope to Prevent Cardiac Implantable Device Infection

ESC

EHRA CONSENSUS PAPER

European Heart Rhythm Association (EHRA) international consensus document on how to prevent, diagnose, and treat cardiac implantable electronic device infections—endorsed by the Heart Rhythm Society (HRS), the Asia Pacific Heart Rhythm Society (APHRS), the Latin American Heart Rhythm Society (LAHRS), International Society for Cardiovascular Infectious Diseases (ISCVID) and the European Society of Clinical Microbiology and Infectious Diseases (ESCMID) in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS)


Circulation: Arrhythmia and Electrophysiology

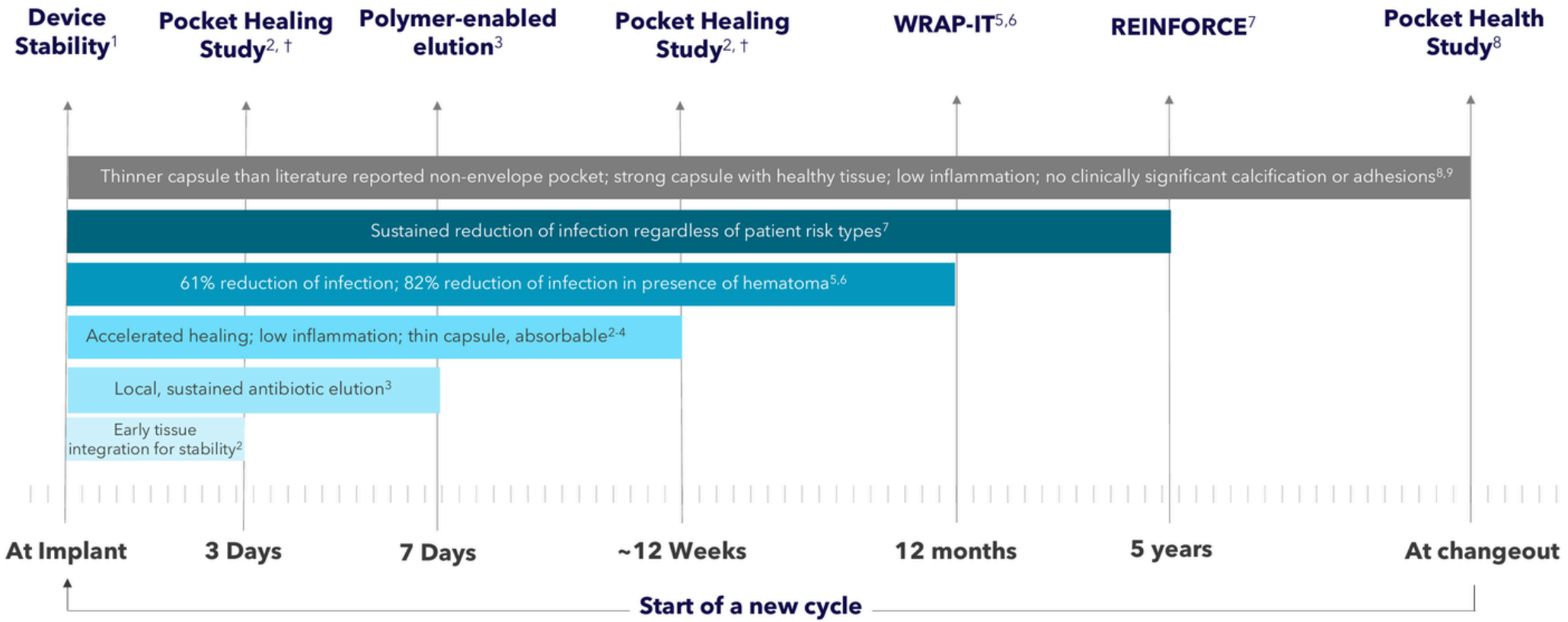
ORIGINAL ARTICLE

Cost-Effectiveness of an Antibacterial Envelope for Cardiac Implantable Electronic Device Infection Prevention in the US Healthcare System From the WRAP-IT Trial

CIED 感染的後果	TYRX 的療效	國際共識建議	成本效益
一年死亡風險 > 3 倍；治療一次感染 5.6 到 8.3 萬美元	主要感染降 40%、囊袋感染降 61%；併發症未增加	EHRA 共識對 WRAP-IT 族群與高風險因子給予正向建議	WRAP-IT 族群 ICER 112,603 美元/QALY，符合 ACC/AHA 門檻

Wilkoff 2020 (e008280)；Tarakji 2019；Blomström-Lundqvist 2020；Wilkoff 2020 (e008503)。表頭可直接點選。

囊袋健康：TYRX 的全程價值



從植入當下的穩定、3 到 7 天的局部溶出、12 週的癒合 (Virmani 2024)，到 12 個月 (WRAP-IT)、5 年 (REINFORCE)，再到下次更換時的囊袋狀態 (Pocket Health Study)。

Take Home Message

1. CIED 感染不常見，但一旦發生，**一年死亡風險超過 3 倍**，花費極高。
2. 七成感染來自 staphylococci，現行全身性抗生素加碼 (PADIT) 壓不下來。
3. TYRX 把兩種抗生素直接送進囊袋：2 小時達 MIC，維持 7 天，9 週吸收。
4. WRAP-IT：主要感染降 **40%**、囊袋感染降 **61%**，併發症不增加；血腫時降 **82%**。
5. 真實世界 (REINFORCE) 在各風險層都有效，NNT 17 到 32。
6. 記得禁忌症：SLE、tetracycline 或 rifampin 過敏、網布過敏、已感染傷口。

關鍵文獻：WRAP-IT、血腫分析、PADIT、REINFORCE、REINFORCE Brady。

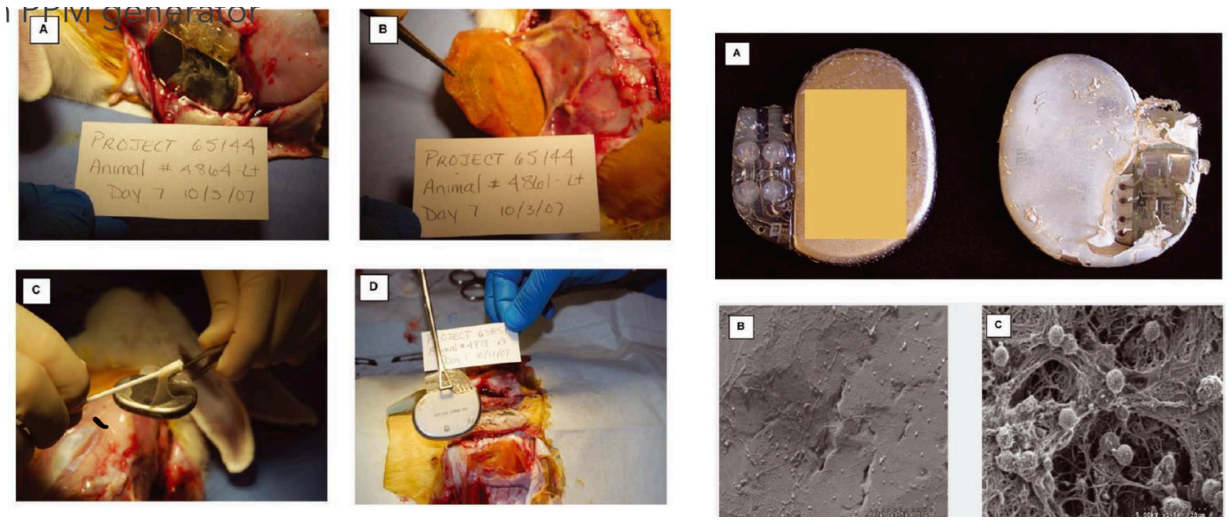
附錄：Micra 無導線心律調節器 (原簡報夾帶)

Micra VR	Micra AV
全球最小的心律調節器	具自動 AV 同步功能
臨床研究植入成功率 > 99%	靜態 85.4%、活動 82.6% AV 同步
併發症較傳統 pacemaker 少 63%	三個月內無需升級
中位電池壽命 13.7 年	中位電池壽命 10.5 年

發表論文 250 篇以上；受訓醫師 9,000 位以上；全球植入 200,000 例以上。原簡報 EMEA Summit 2024，為 Medtronic 行銷數據，本頁未逐一查證。

備用：兔子感染模型

- 兔子雙側皮下植入 PPM generator，一側加第一代 TYRX
- 囊袋分別接種 S. capitis、E. coli、S. epidermidis、Acinetobacter baumannii
- 關閉囊袋 7 天
- **對照側全部感染**，TYRX 側乾淨無生物膜



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縮寫對照表

縮寫	全名	中文
CIED	Cardiac Implantable Electronic Device	心臟植入式電子裝置
CRT-D / CRT-P	Cardiac Resynchronization Therapy Defibrillator / Pacemaker	心臟再同步治療去顫器 / 節律器
ICD	Implantable Cardioverter Defibrillator	植入式去顫器
MIC	Minimum Inhibitory Concentration	最低抑菌濃度
NNT	Number Needed to Treat	需治療人數
PADIT	Prevention of Arrhythmia Device Infection Trial	心律裝置感染預防試驗
WRAP-IT	Worldwide Randomized Antibiotic Envelope Infection Prevention Trial	全球隨機抗生素包膜感染預防試驗
ECM	Extracellular Matrix	細胞外基質

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Q & A

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